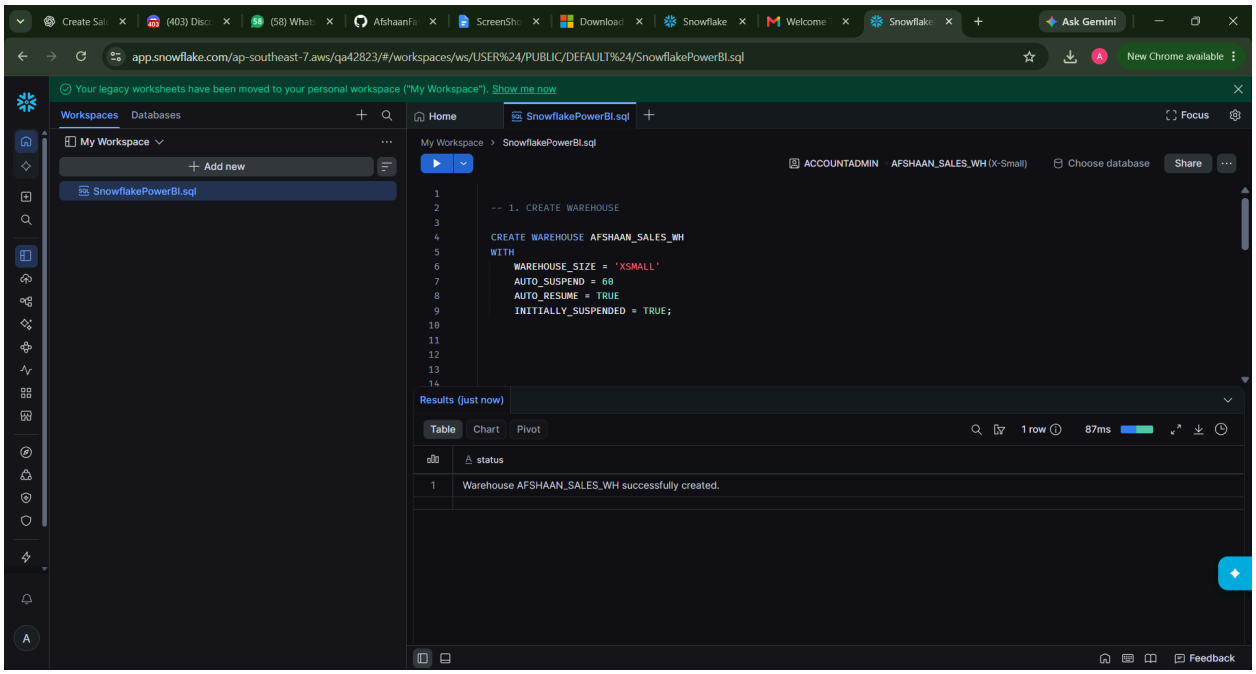
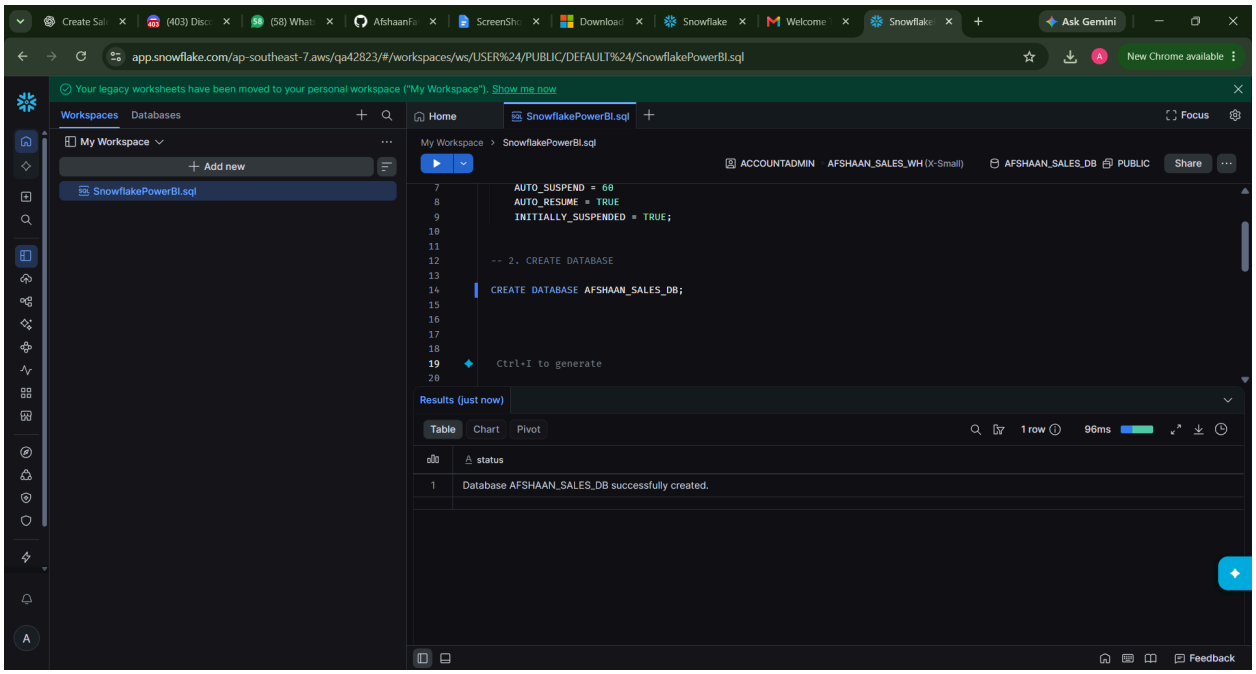
SNOWFLAKE- POWERBI INTEGRATION

SNOWFLAKE

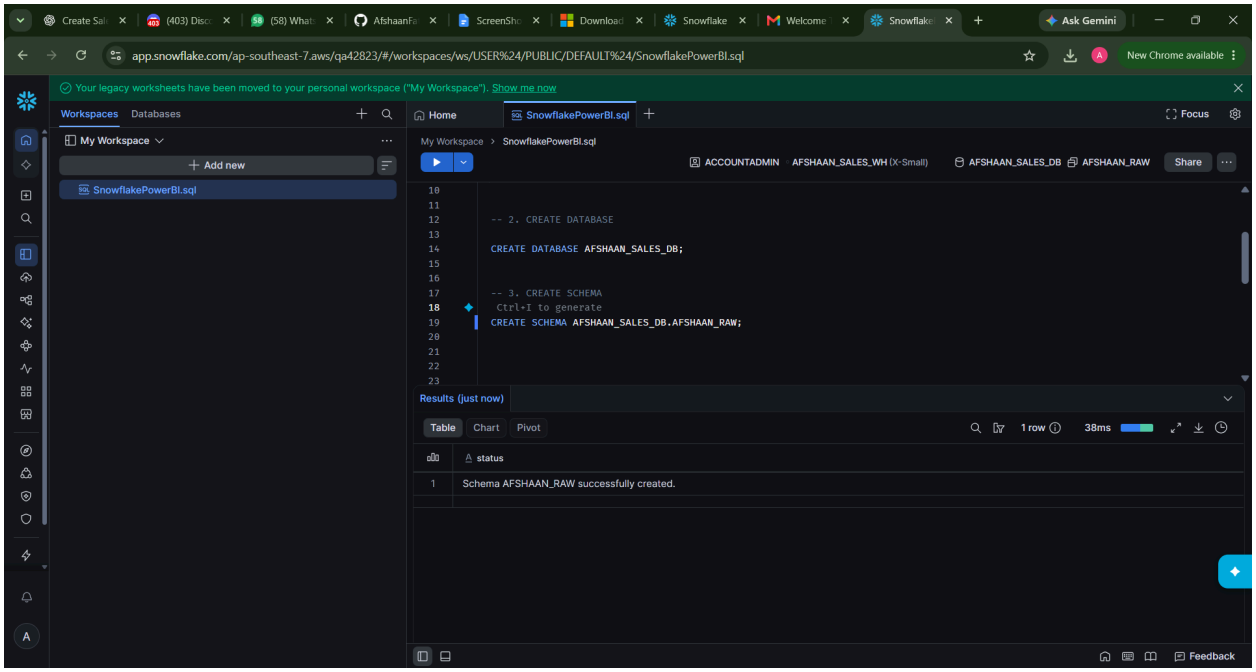
Create Warehouse



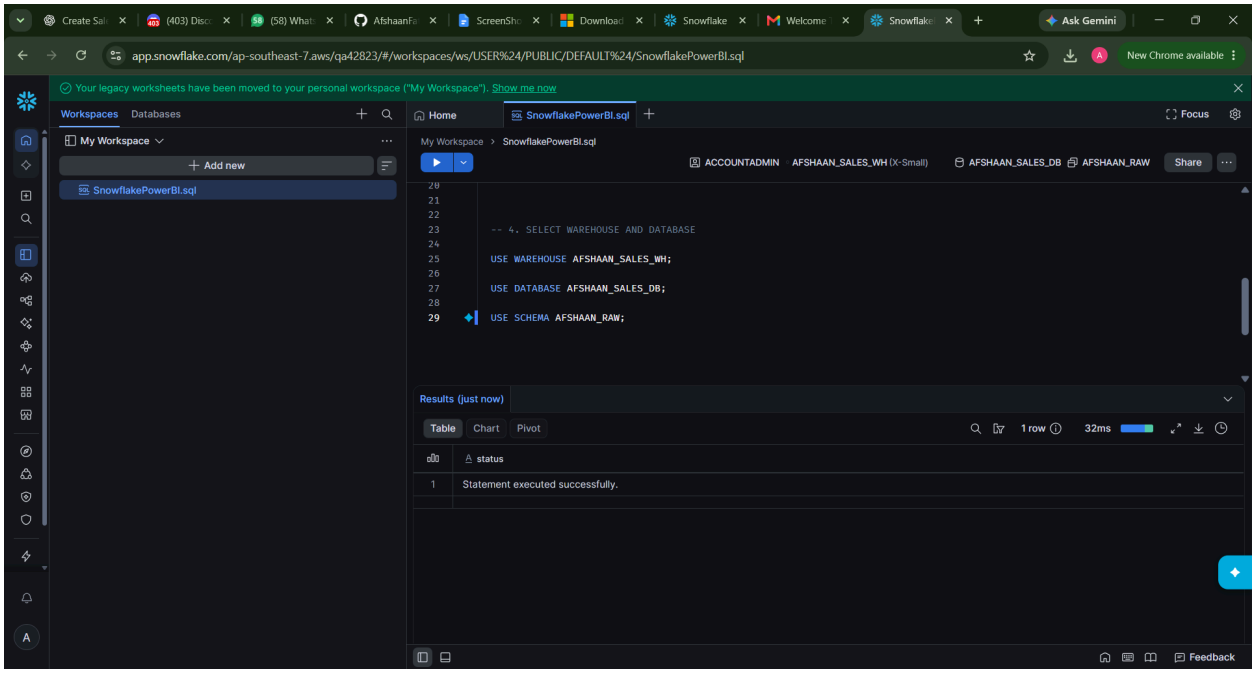
Create Database



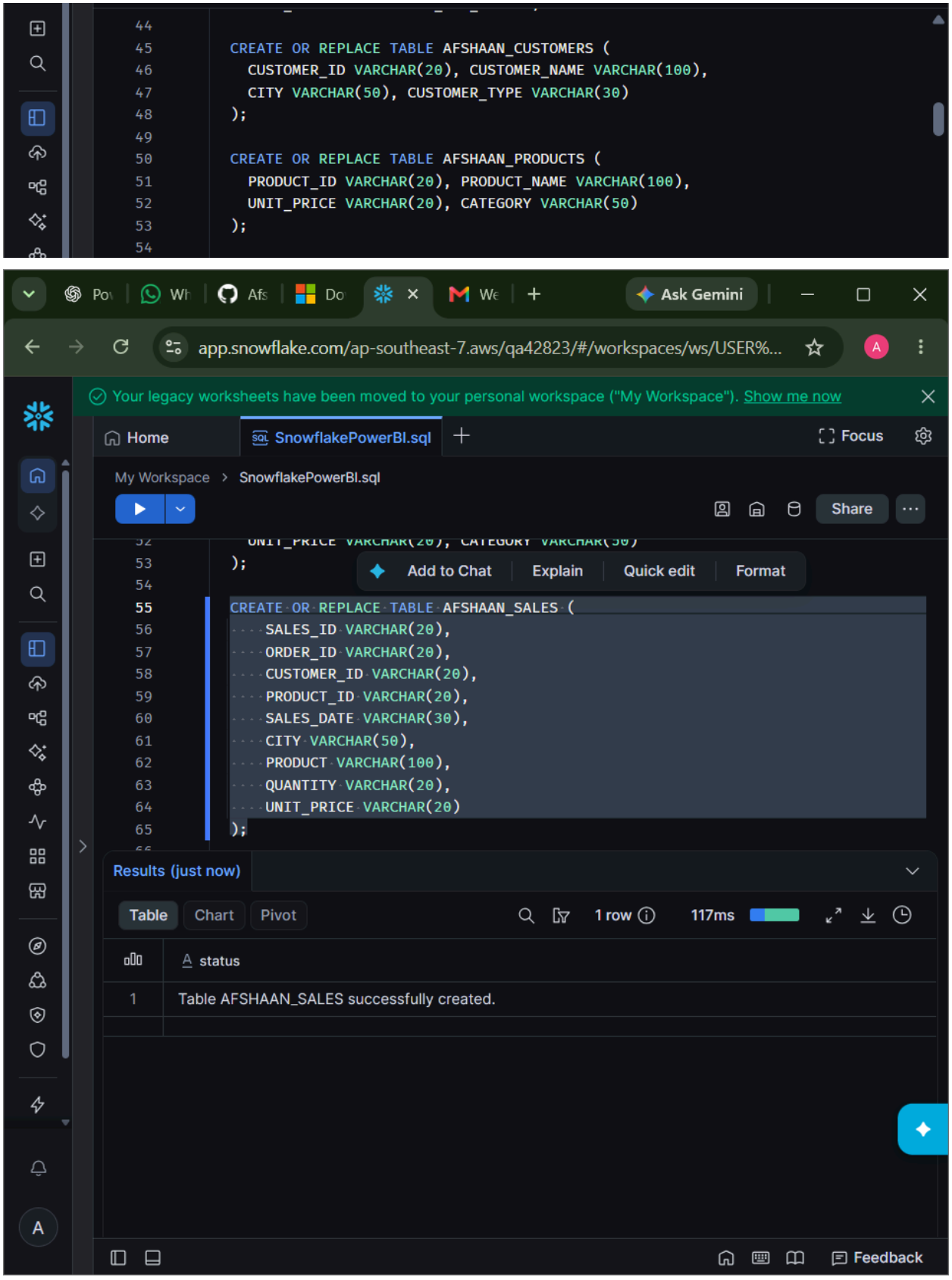
Create Schema



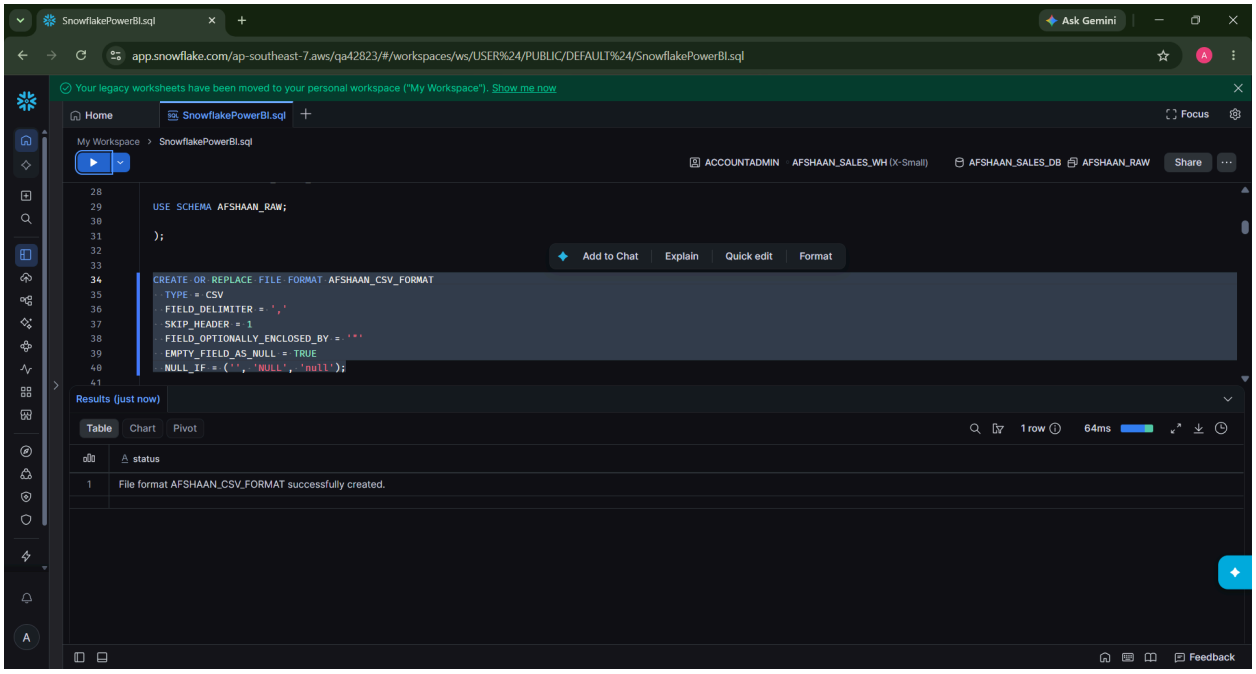
Select Warehouse and Db



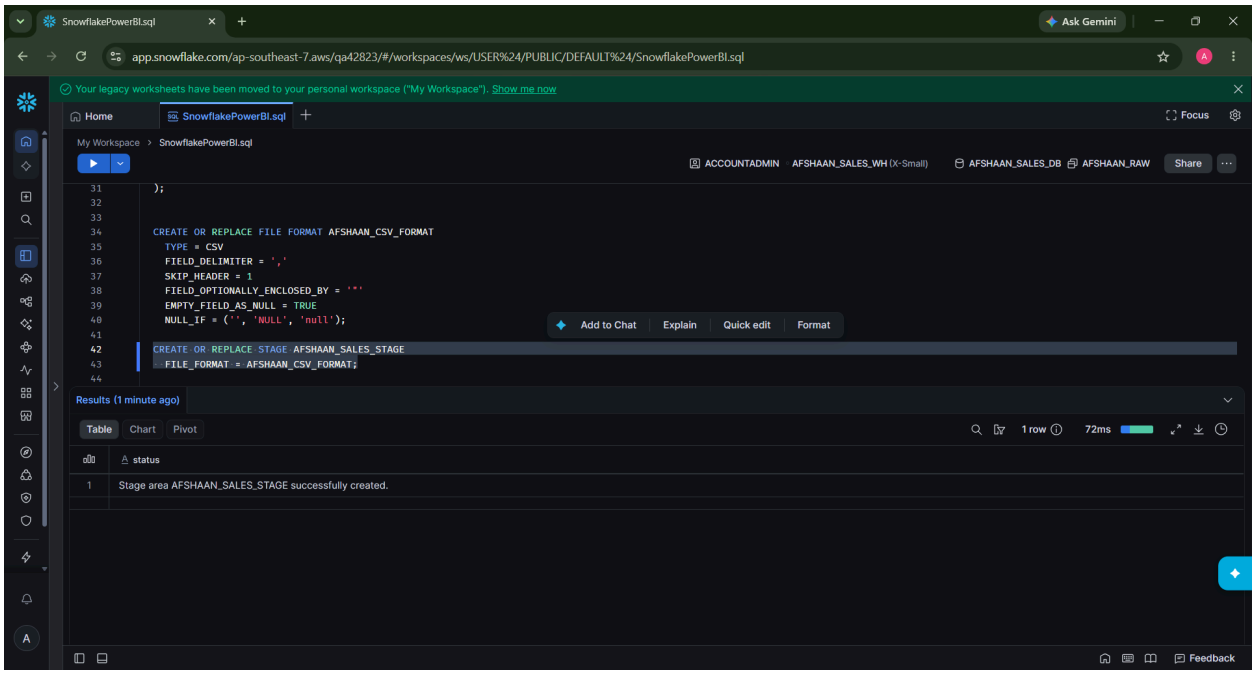
Create tables



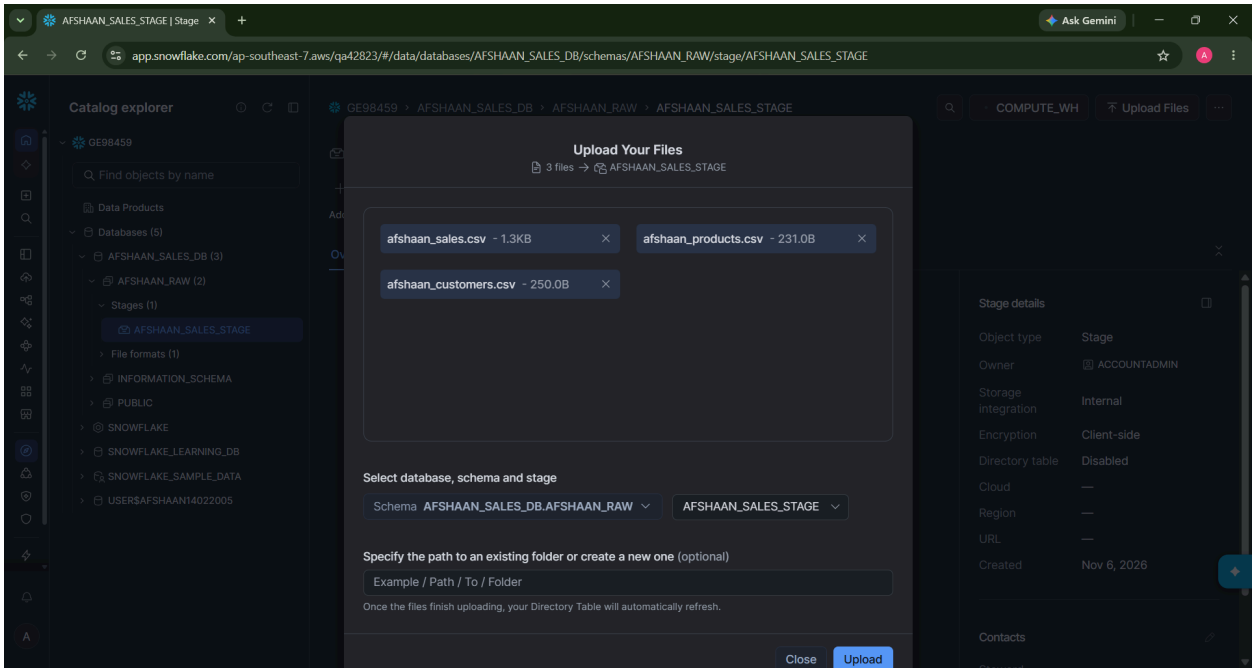
Create File Format



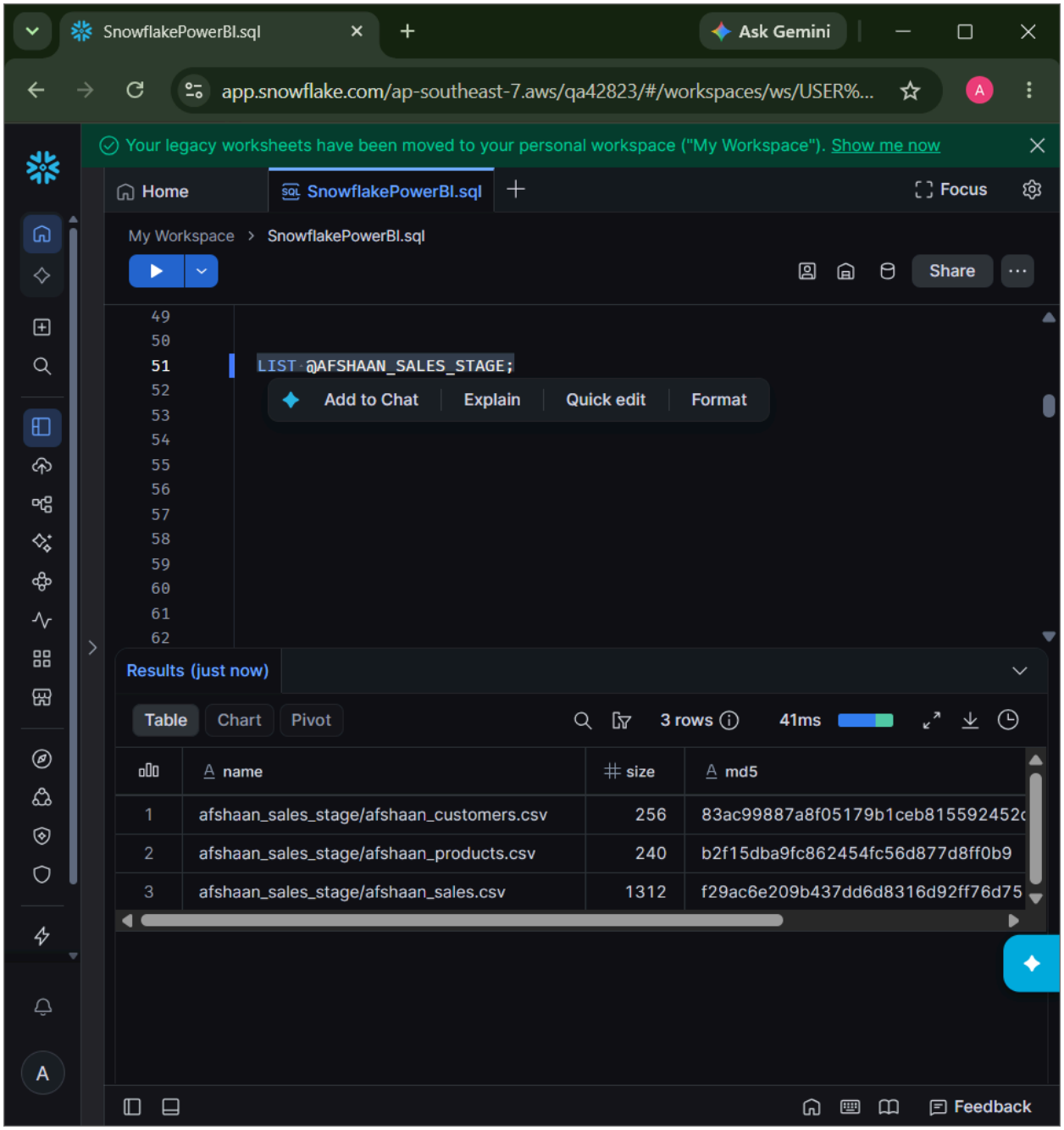
Create Stage



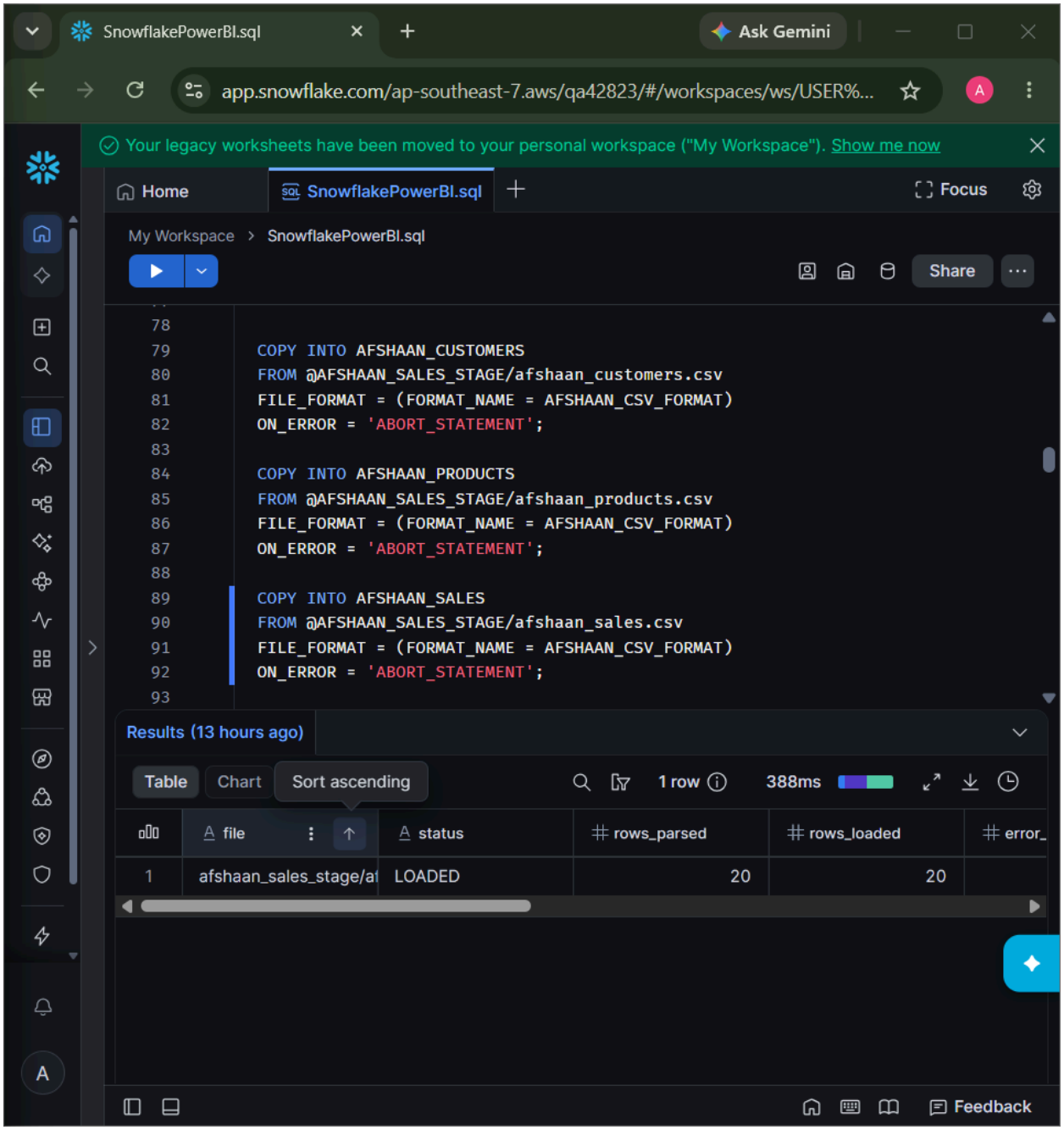
Upload the files into the stage



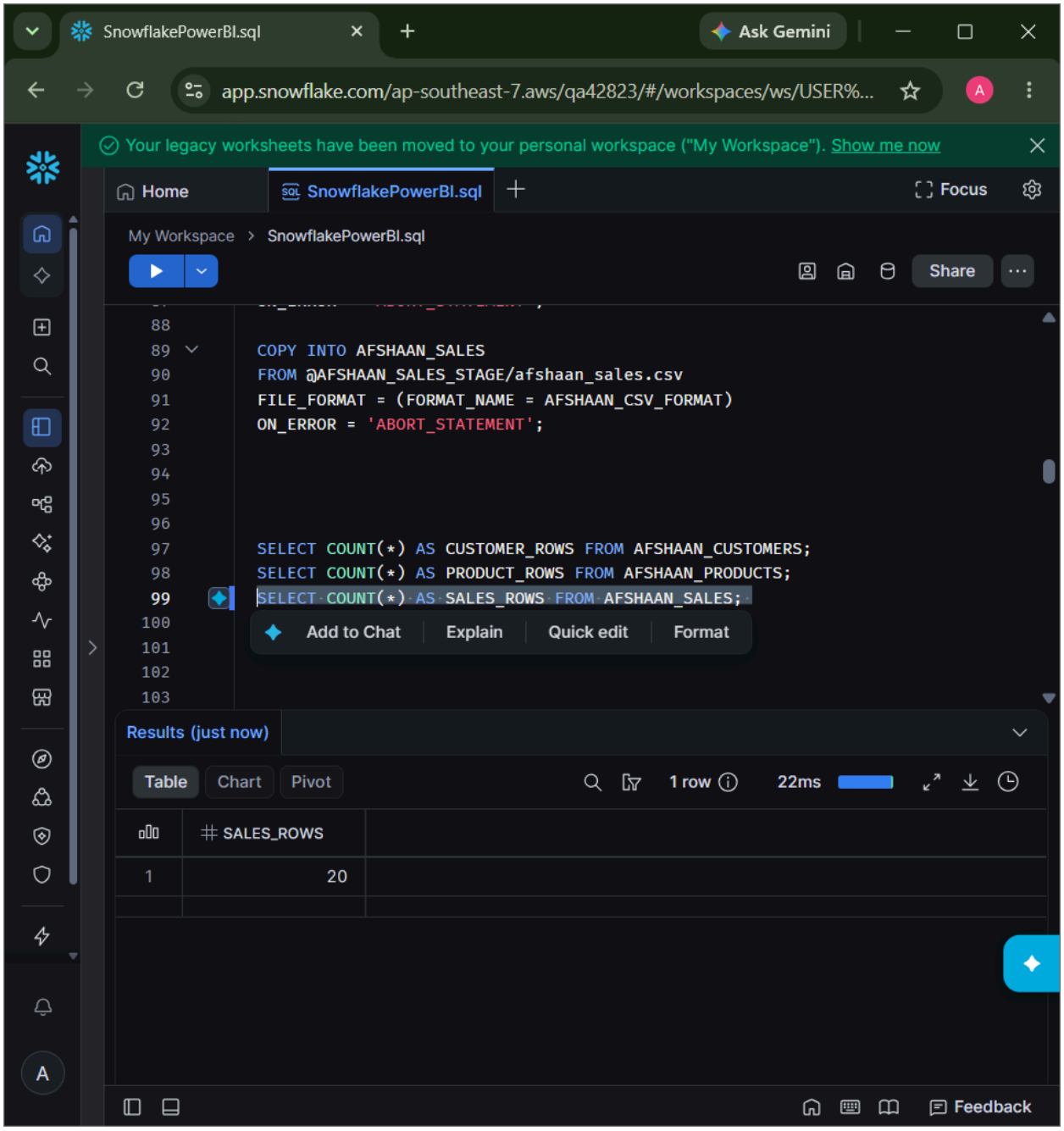
Verify the uploads by listing the stage

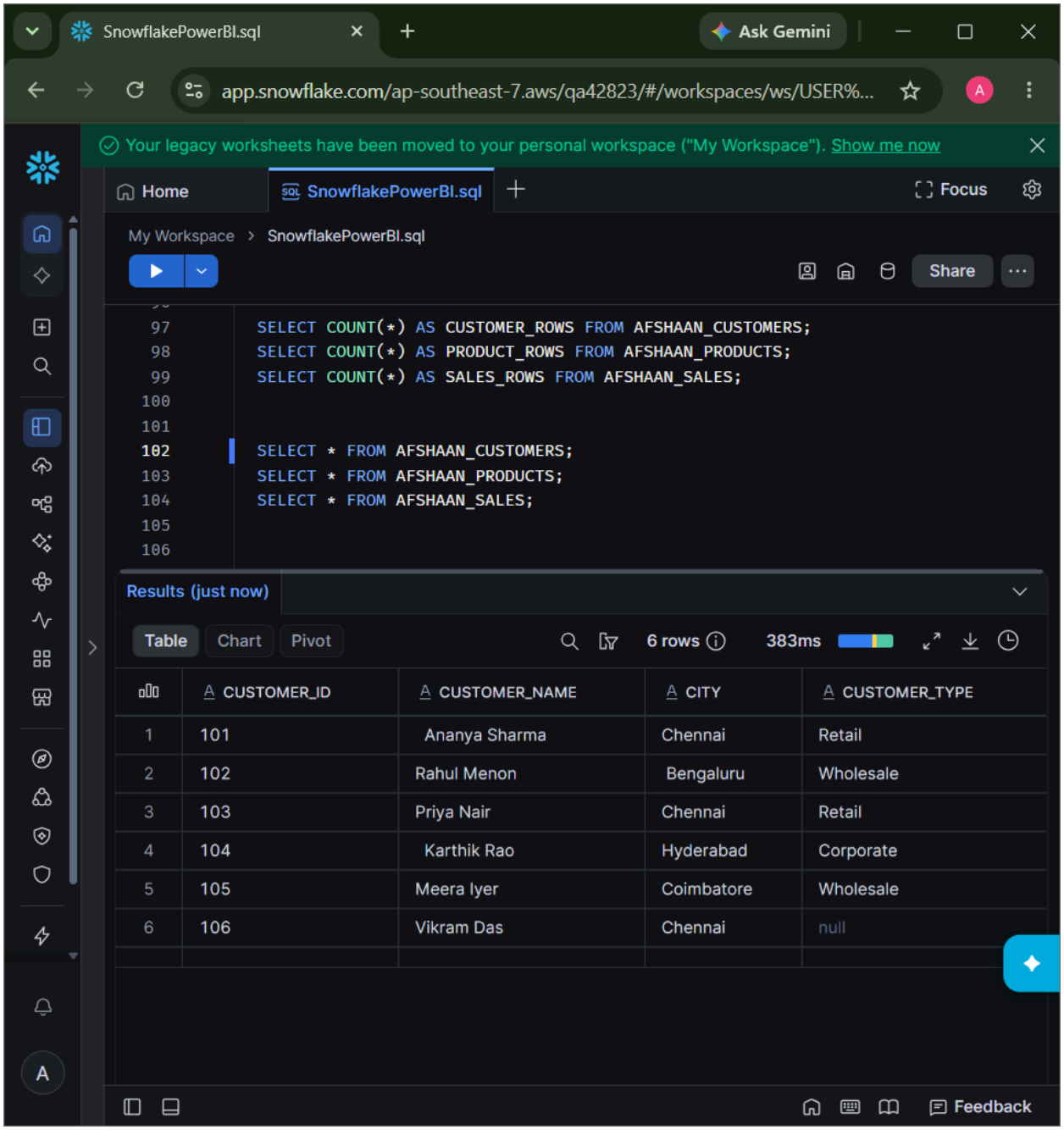


Copy the staged tables into the Table using Copy Into

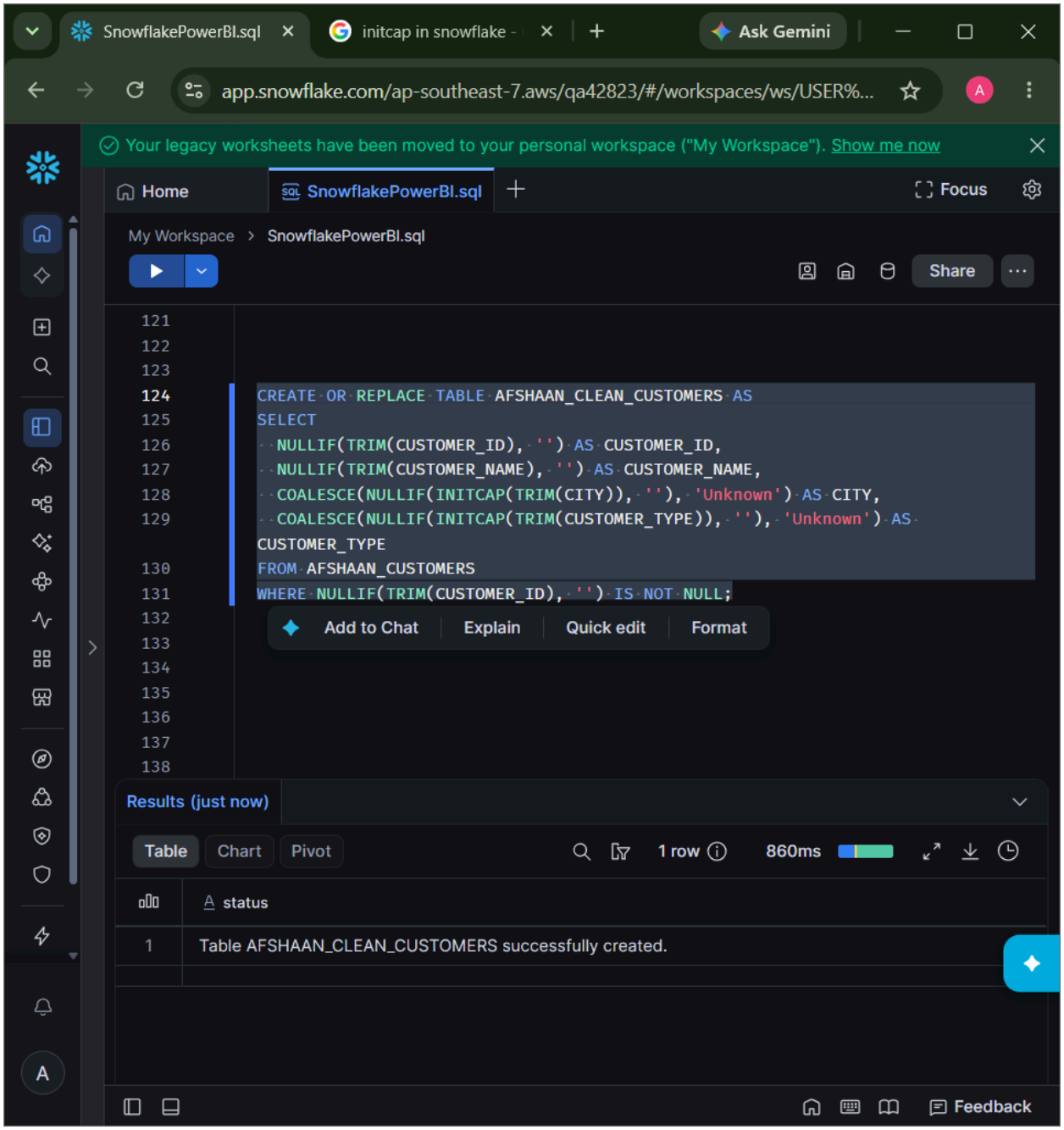


Verify the tables

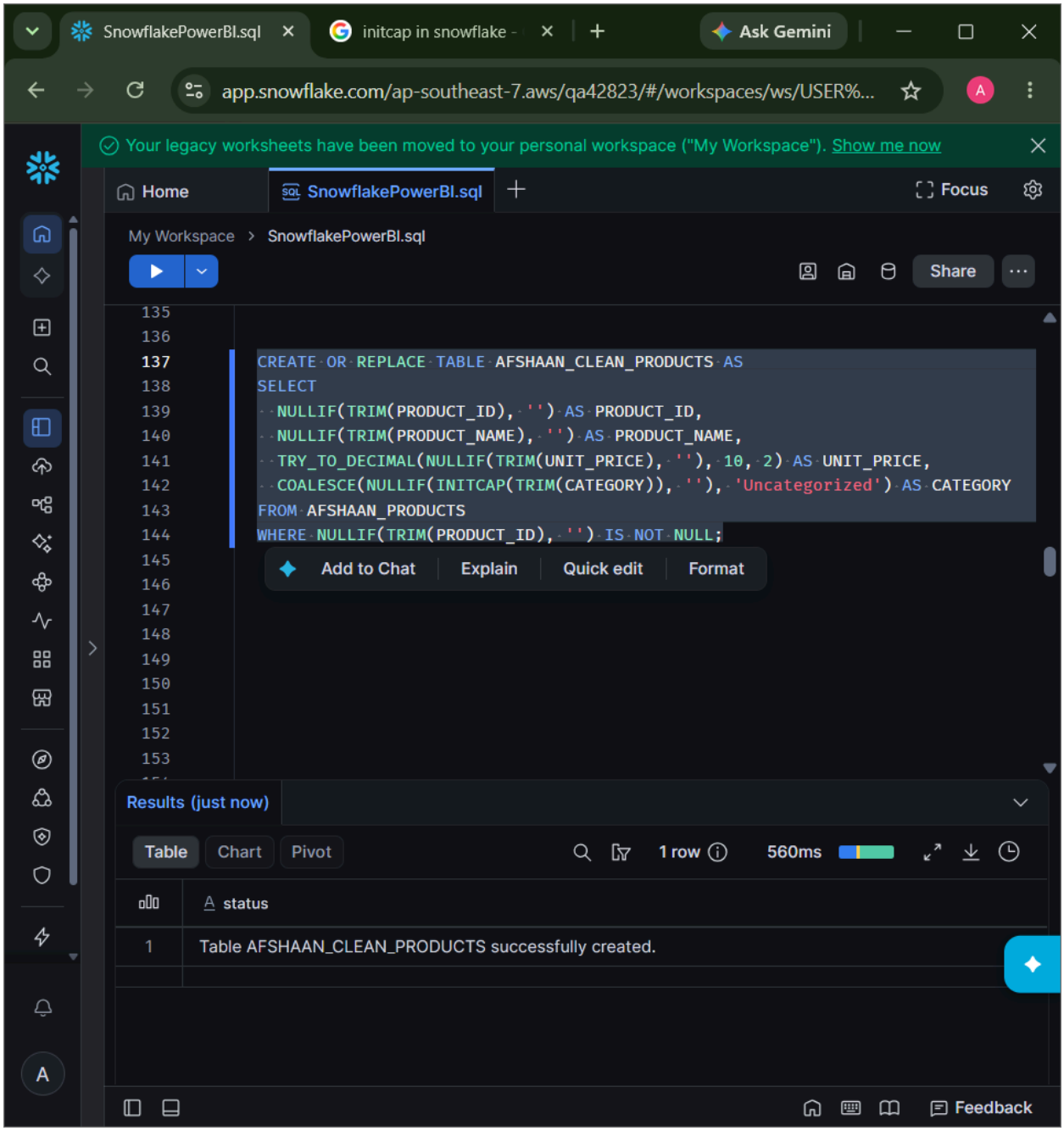

SnowflakePowerBI.sql
Ask Gemini
app.snowflake.com/ap-southeast-7.aws/qa42823/#/workspaces/ws/USER%...
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Home
SnowflakePowerBI.sql
Focus
My Workspace > SnowflakePowerBI.sql
Share
88
89 COPY INTO AFSHAAN_SALES
90 FROM @AFSHAAN_SALES_STAGE/afshaan_sales.csv
91 FILE_FORMAT = (FORMAT_NAME = AFSHAAN_CSV_FORMAT)
92 ON_ERROR = 'ABORT_STATEMENT';
93
94
95
96
97 SELECT COUNT(*) AS CUSTOMER_ROWS FROM AFSHAAN_CUSTOMERS;
98 SELECT COUNT(*) AS PRODUCT_ROWS FROM AFSHAAN_PRODUCTS;
99 SELECT COUNT(*) AS SALES_ROWS FROM AFSHAAN_SALES;
100
101
102
103
Add to Chat
Explain
Quick edit
Format
Results (just now)
Table
Chart
Pivot
1 row
22ms
SALES_ROWS
1
20
Feedback

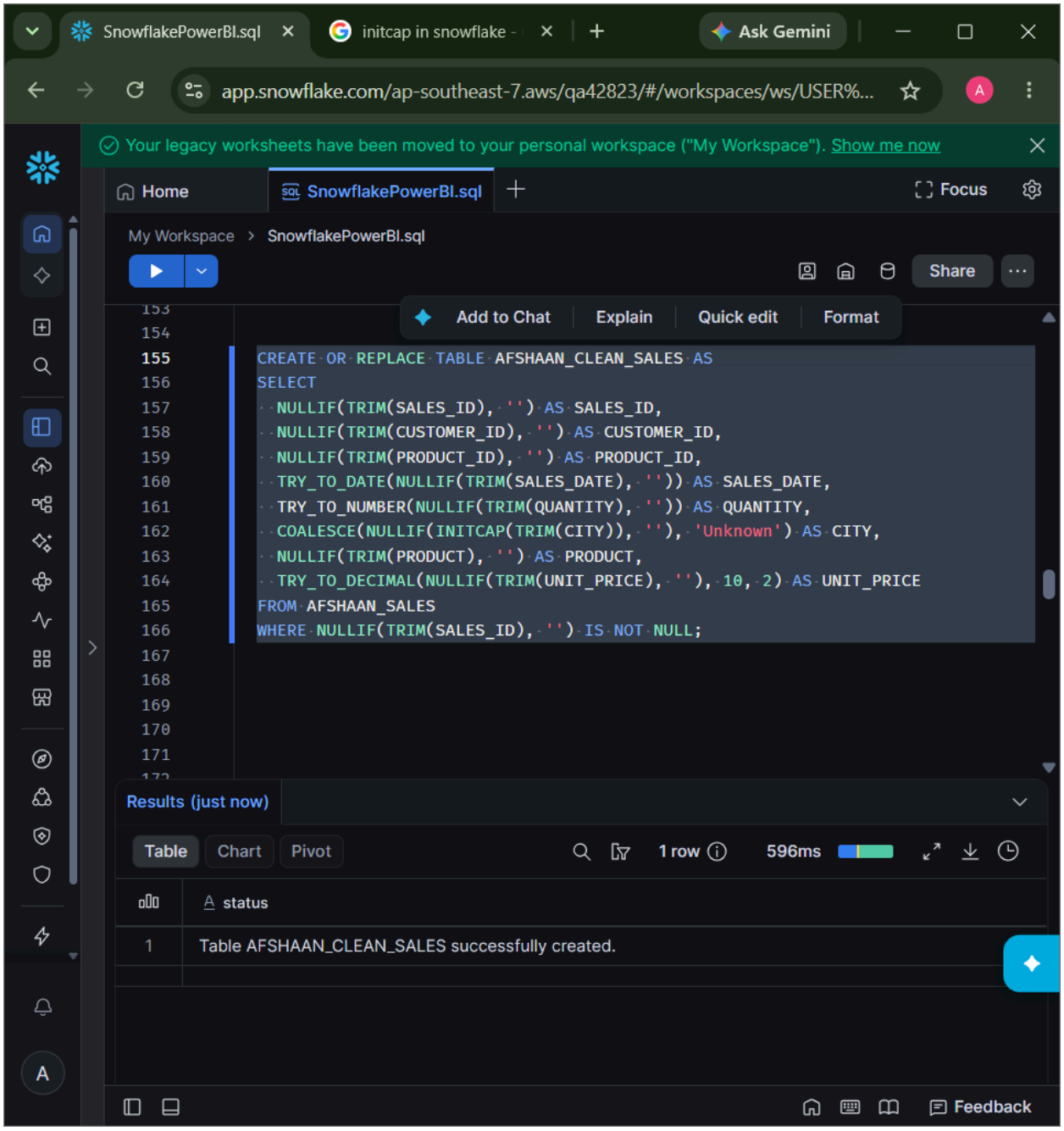


Cleaned Tables:
AFSHAAN_CLEAN_CUSTOMERS

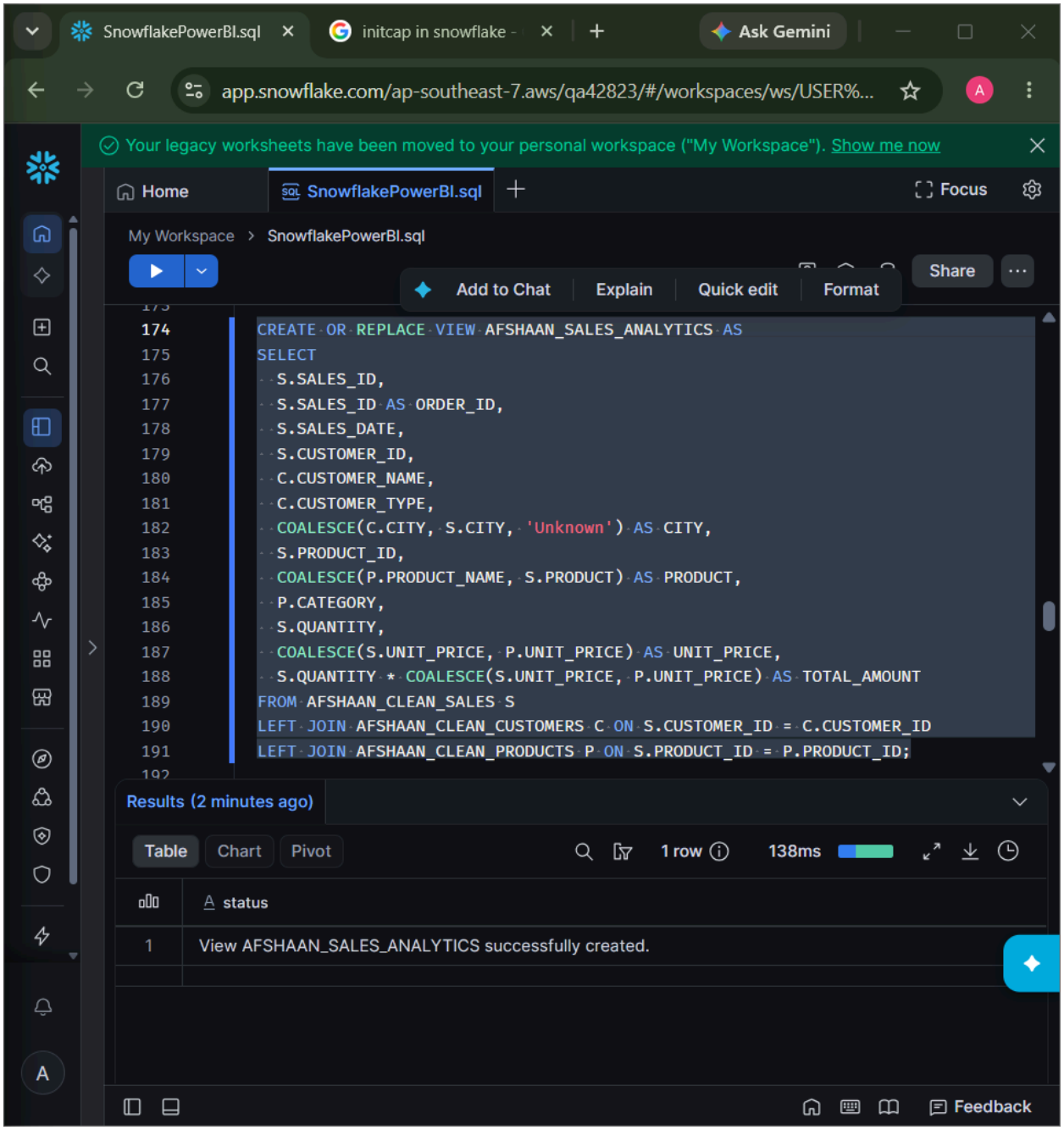


AFSHAAN_CLEAN_PRODUCTS


SnowflakePowerBI.sql
initcap in snowflake -
Ask Gemini
app.snowflake.com/ap-southeast-7.aws/qa42823/#/workspaces/ws/USER%...
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Home
SnowflakePowerBI.sql
Focus
My Workspace > SnowflakePowerBI.sql
Share
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
CREATE OR REPLACE TABLE AFSHAAN_CLEAN_PRODUCTS AS
SELECT
NULLIF(TRIM(PRODUCT_ID), '') AS PRODUCT_ID,
NULLIF(TRIM(PRODUCT_NAME), '') AS PRODUCT_NAME,
TRY_TO_DECIMAL(NULLIF(TRIM(UNIT_PRICE), ''), 10, 2) AS UNIT_PRICE,
COALESCE(NULLIF(INITCAP(TRIM(CATEGORY)), ''), 'Uncategorized') AS CATEGORY
FROM AFSHAAN_PRODUCTS
WHERE NULLIF(TRIM(PRODUCT_ID), '') IS NOT NULL;
Add to Chat
Explain
Quick edit
Format
Results (just now)
Table
Chart
Pivot
1 row
560ms
status
1
Table AFSHAAN_CLEAN_PRODUCTS successfully created.
Feedback

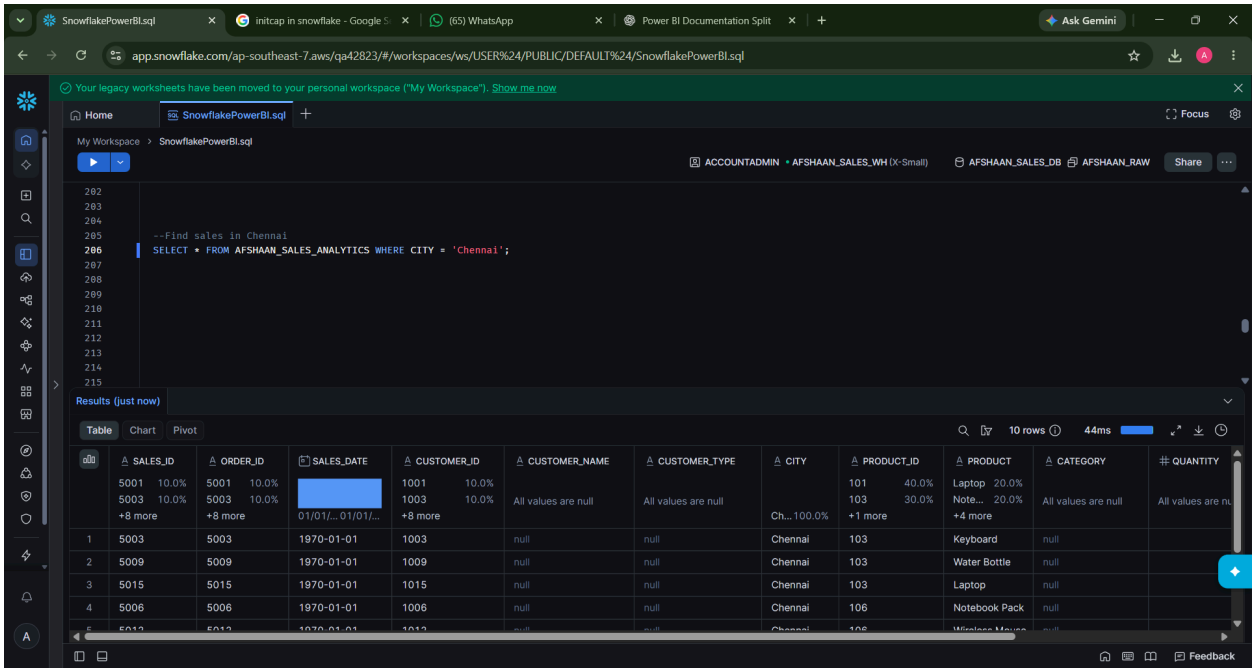


Creating a view for easier queries

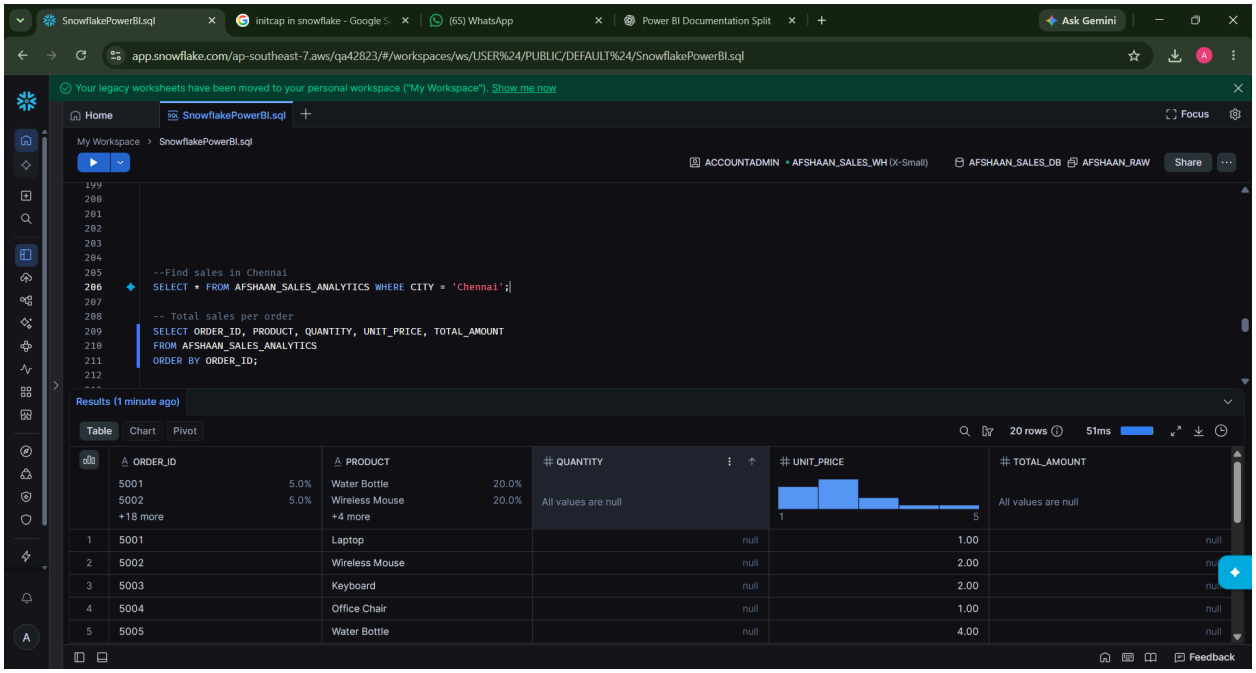


Analytical queries

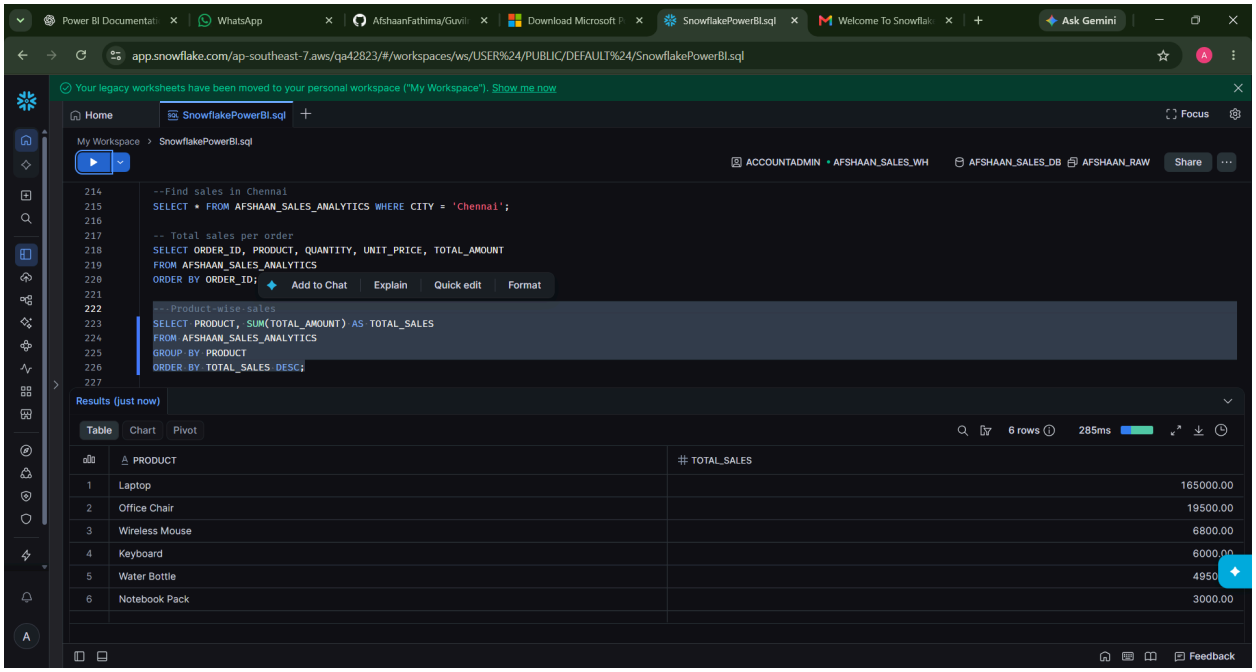
Find sales in Chennai



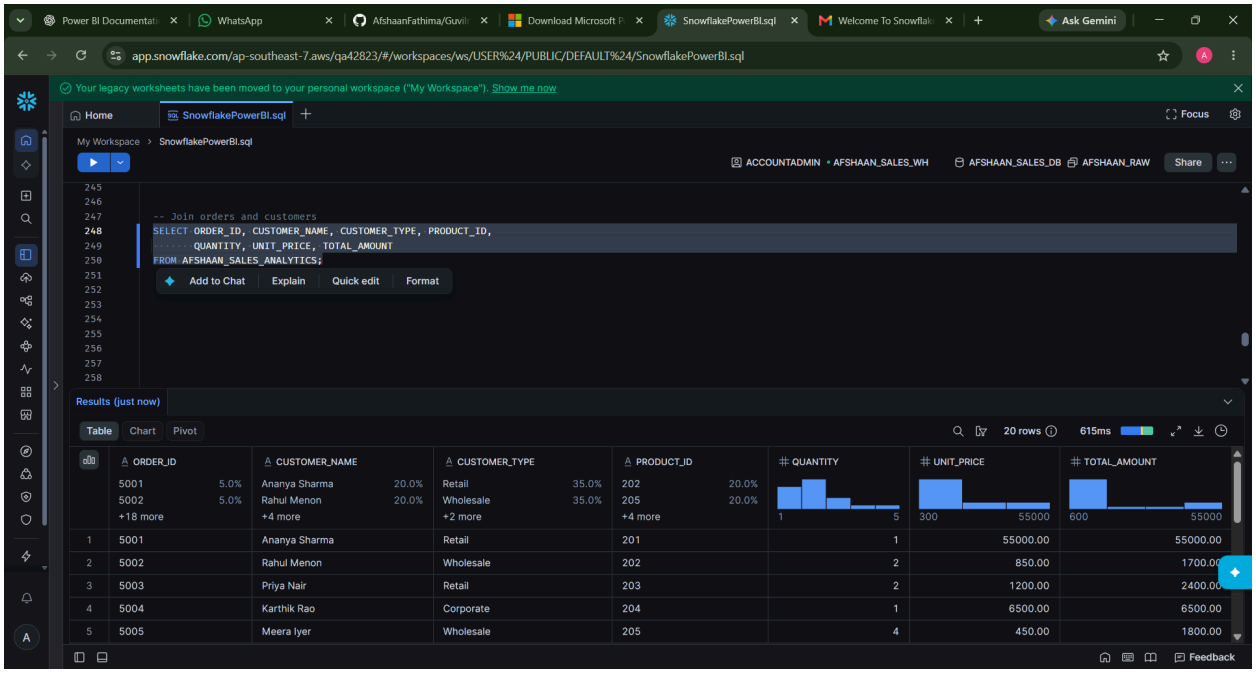
Total sales per order



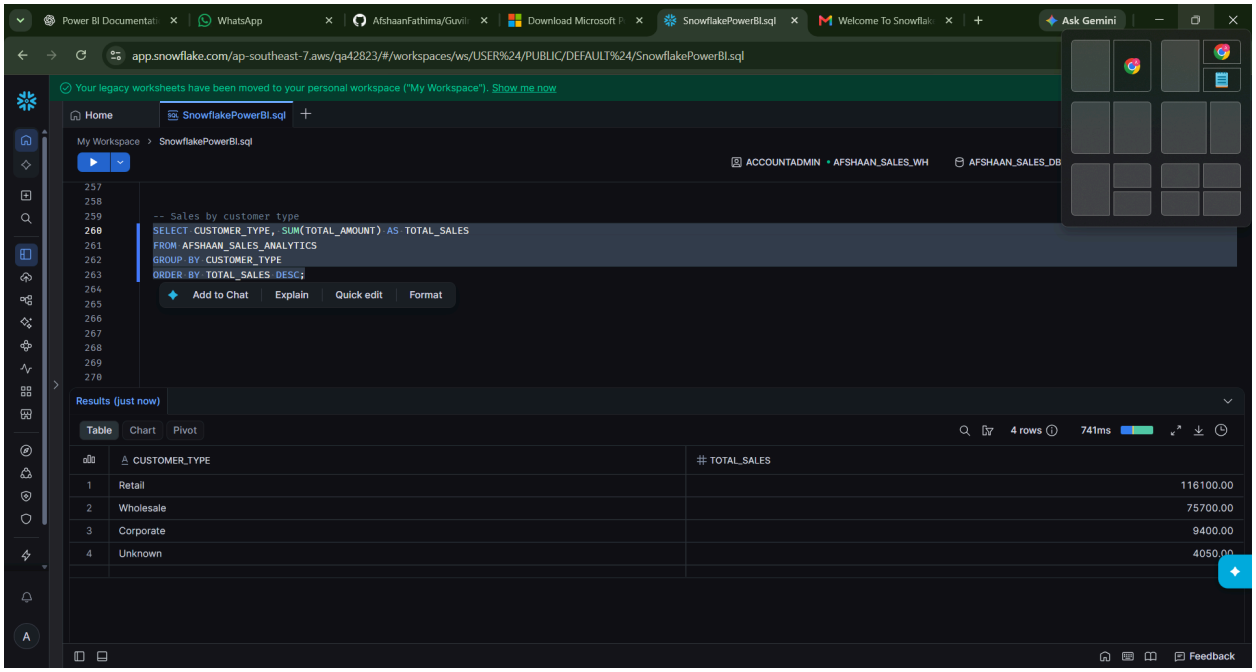
Product-wise sales



Join orders and customers

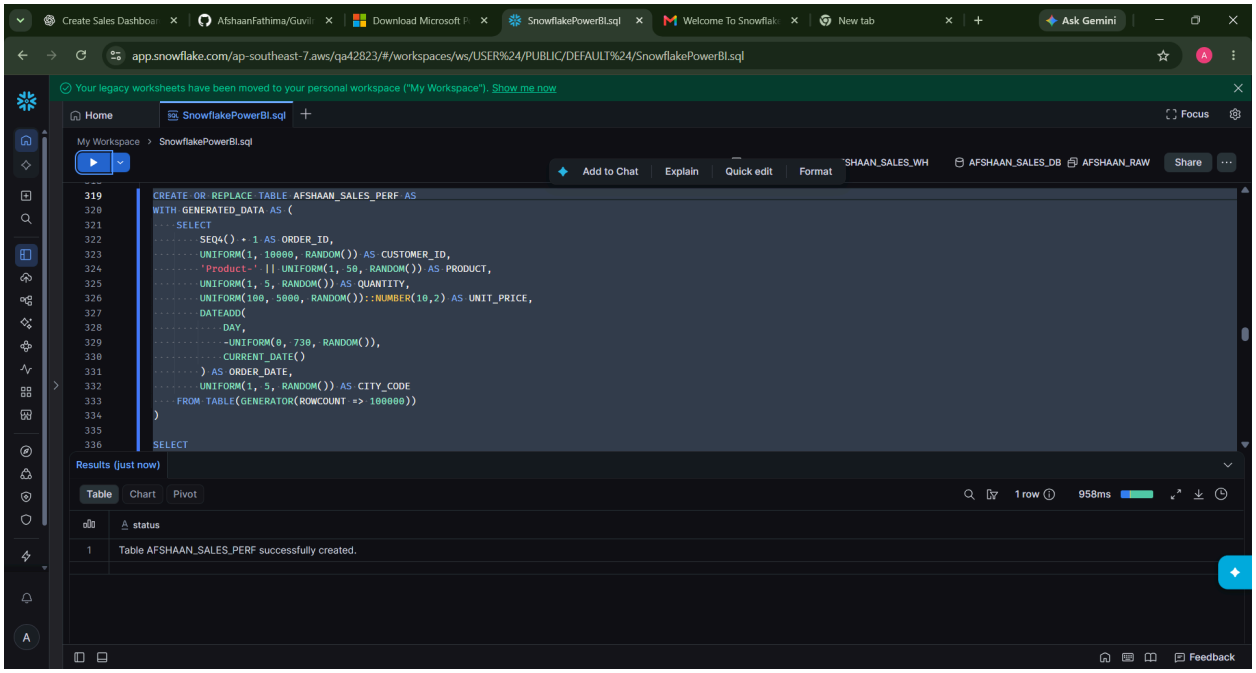


Sales by customer type

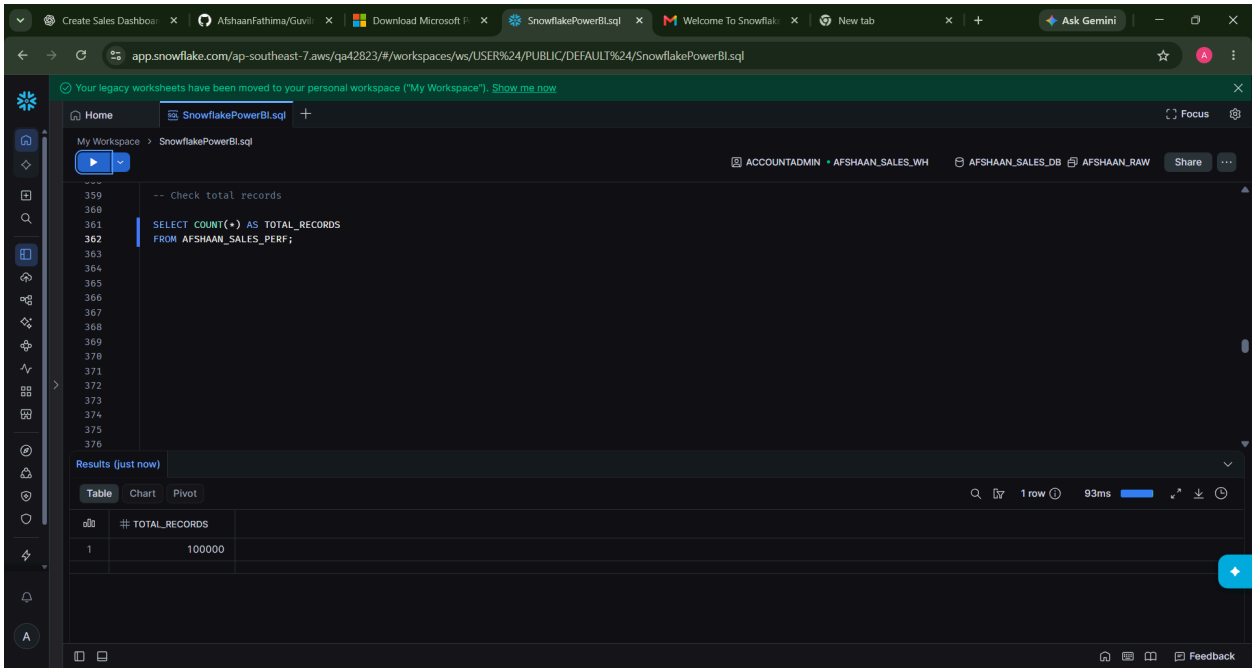


QUERY OPTIMIZATION

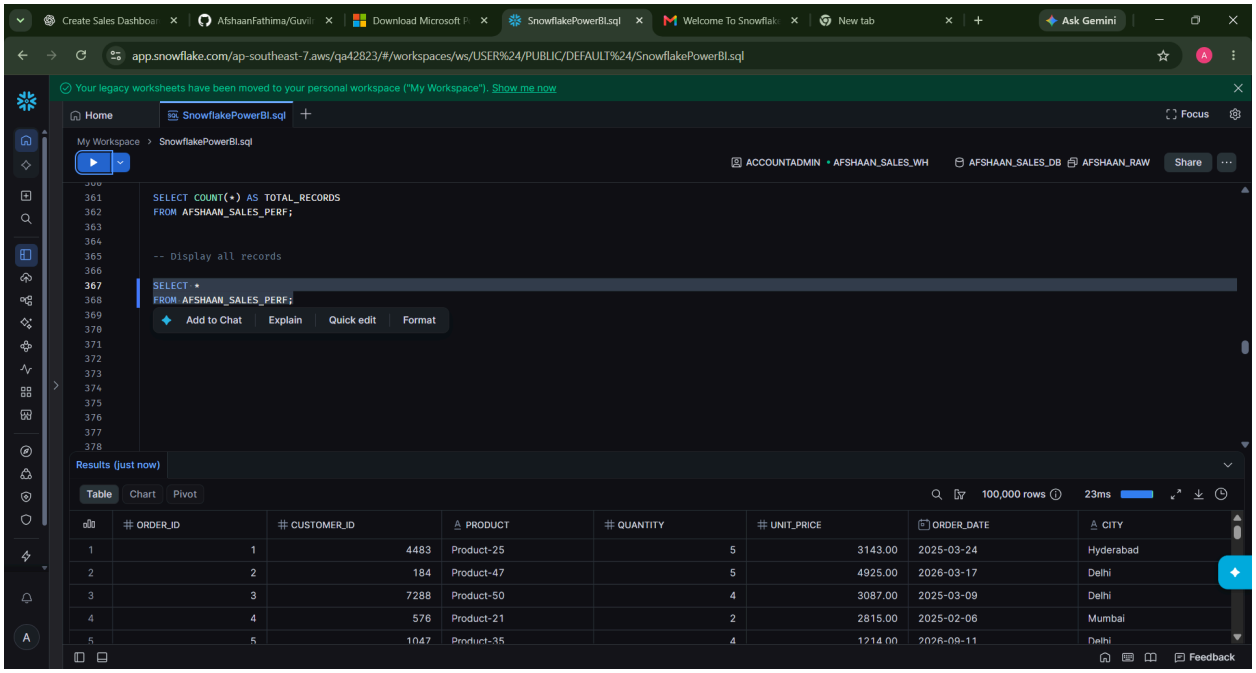
GENERATE 100000 RECORDS



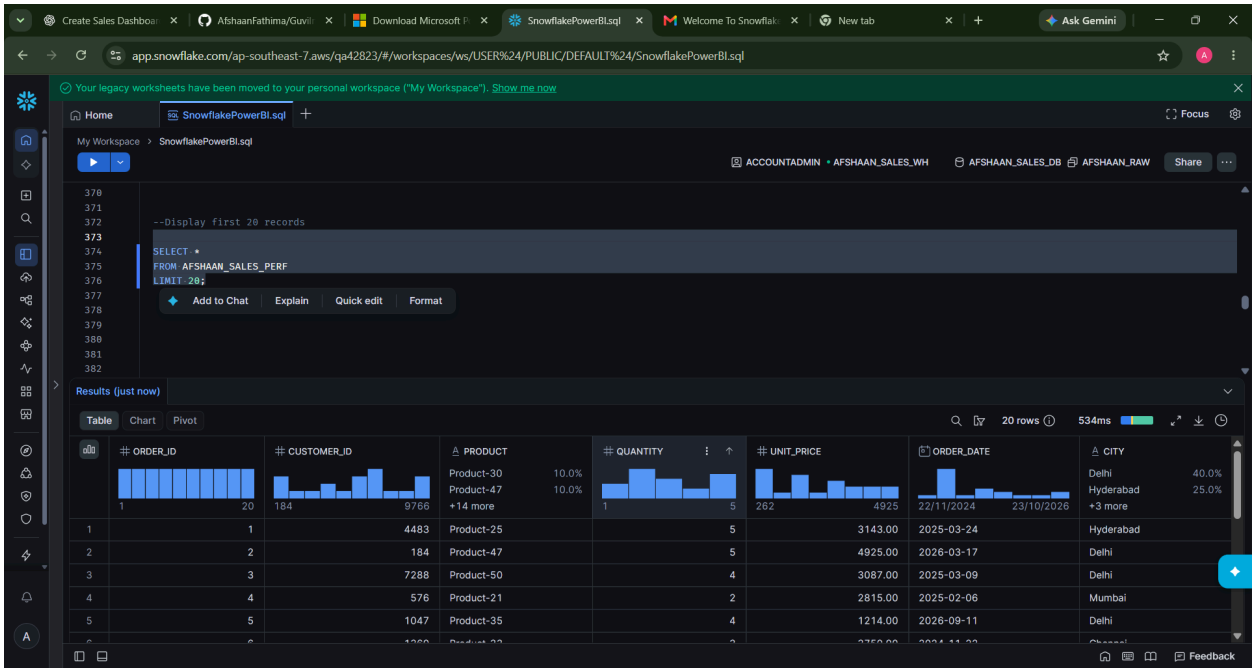
Check total records



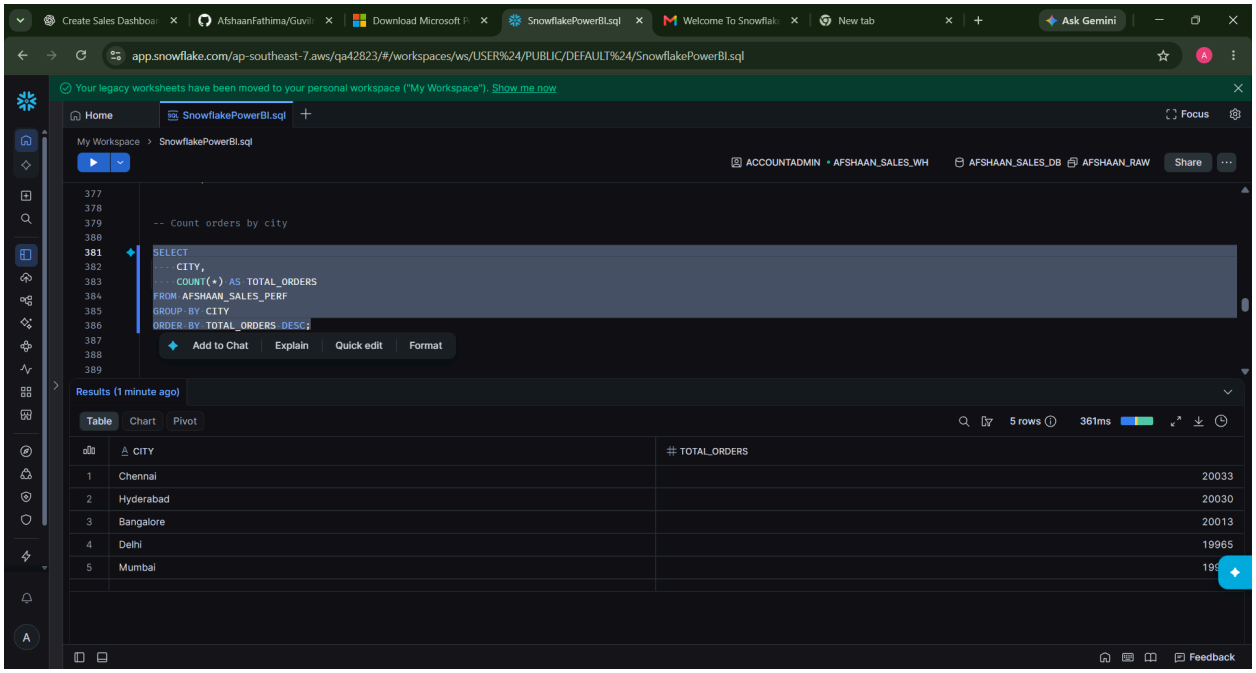
Display all records



Display first 20 records



Count orders by city

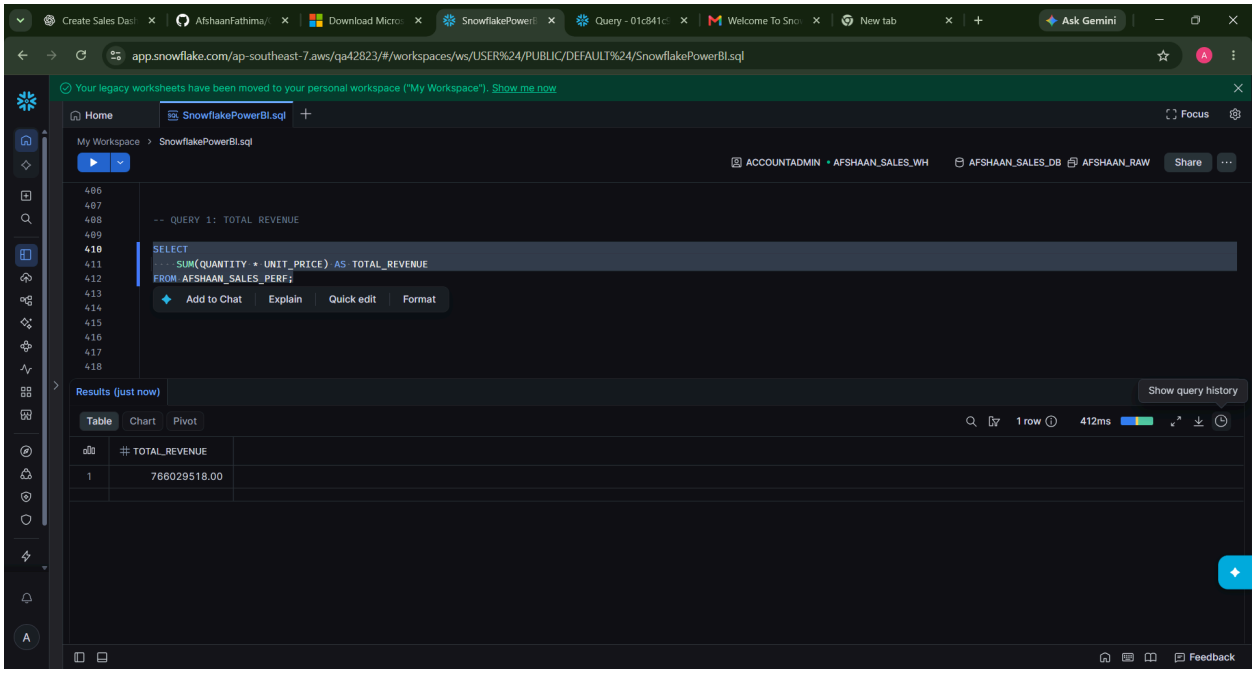


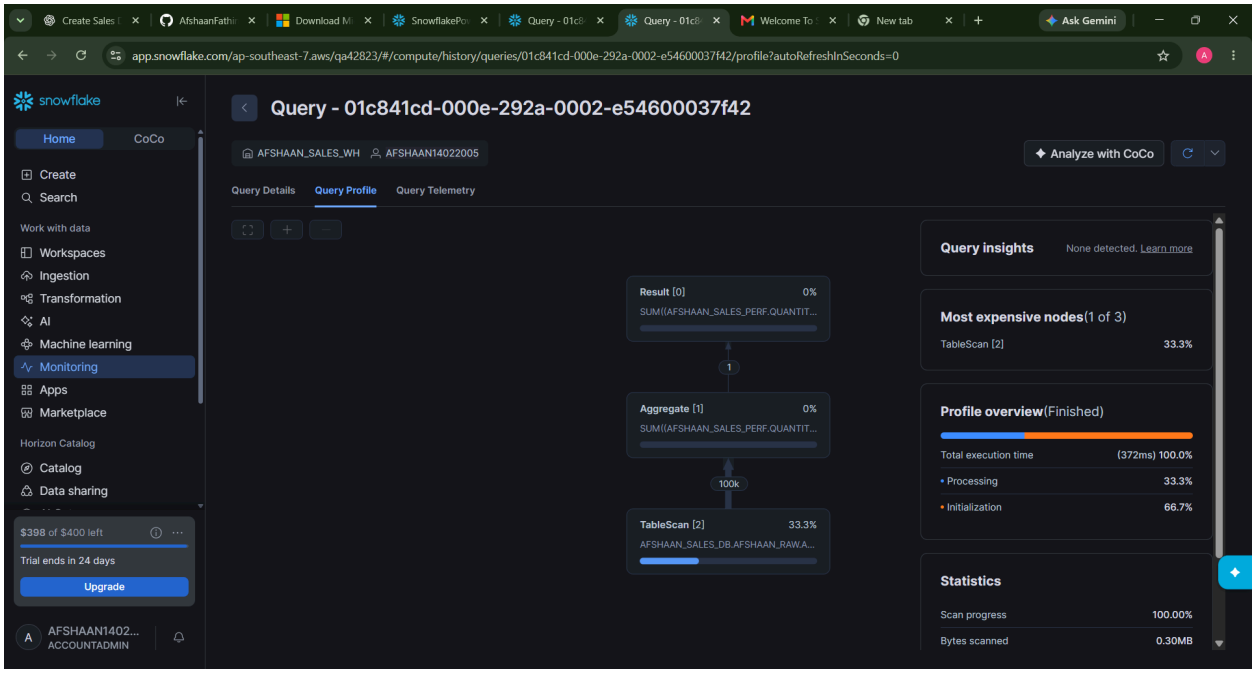
QUERIES- Comparison with Optimized queries

QUERY 1: TOTAL REVENUE

BASELINE

```
SELECT
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE
FROM AFSHAAN_SALES_PERF;
```

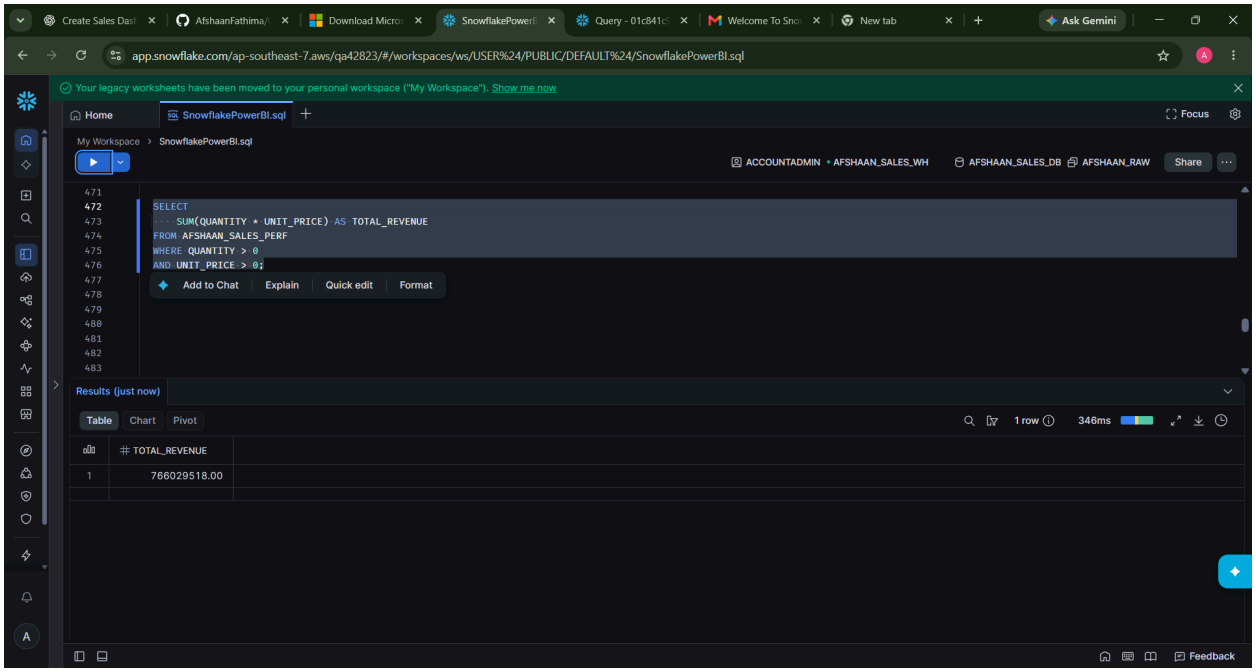


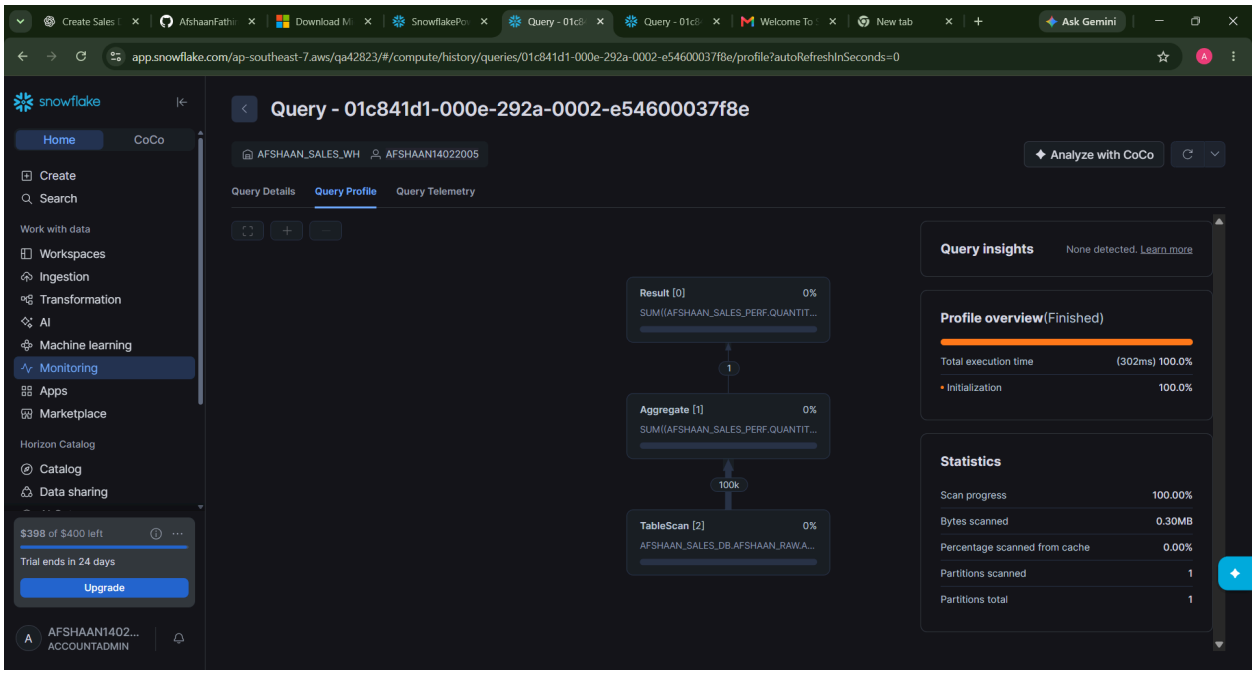


OPTIMIZED

Apply filtering to exclude invalid values

```
SELECT
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE
FROM AFSHAAN_SALES_PERF
WHERE QUANTITY > 0
AND UNIT_PRICE > 0;
```



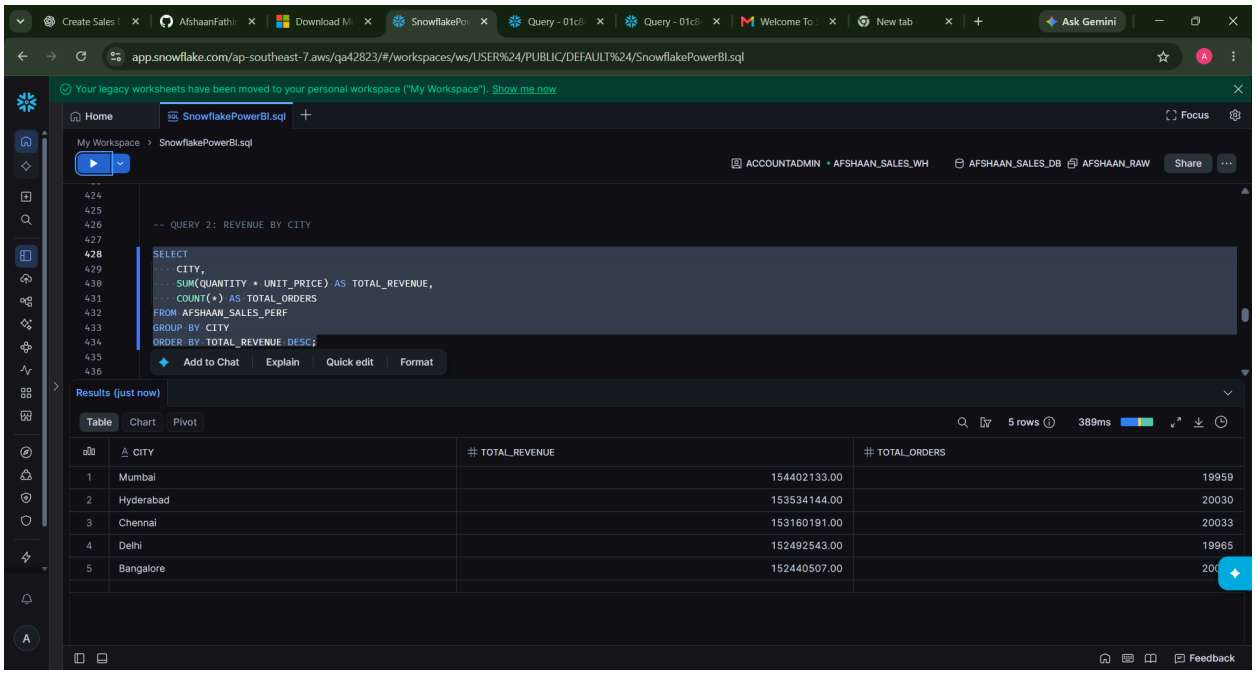


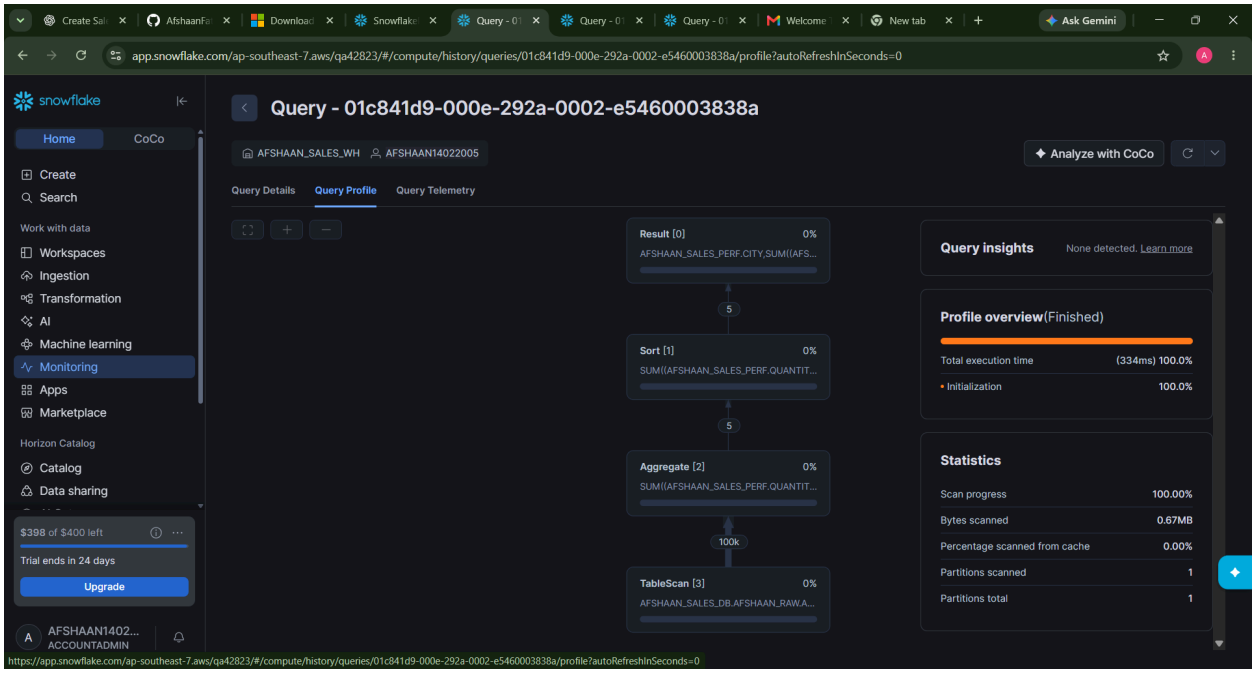
QUERY 2: REVENUE BY CITY

BASELINE

```
SELECT
    CITY,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE,
    COUNT(*) AS TOTAL_ORDERS
FROM AFSHAAN_SALES_PERF
GROUP BY CITY
```

ORDER BY TOTAL_REVENUE DESC;



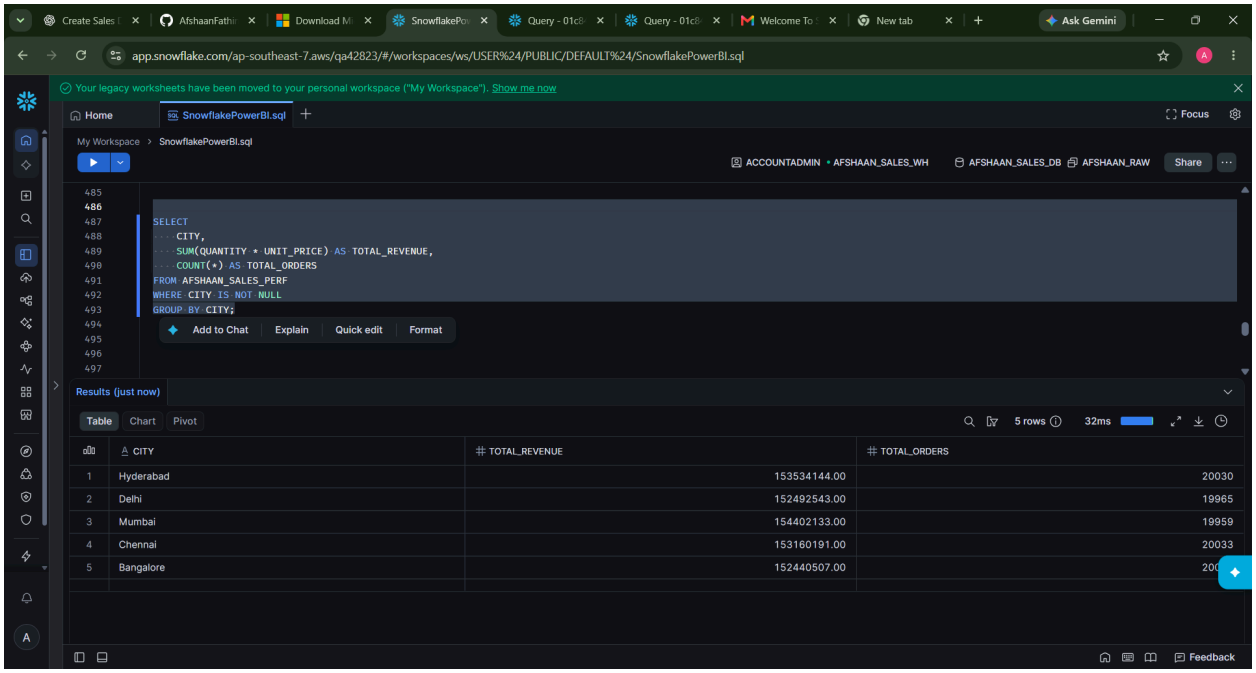


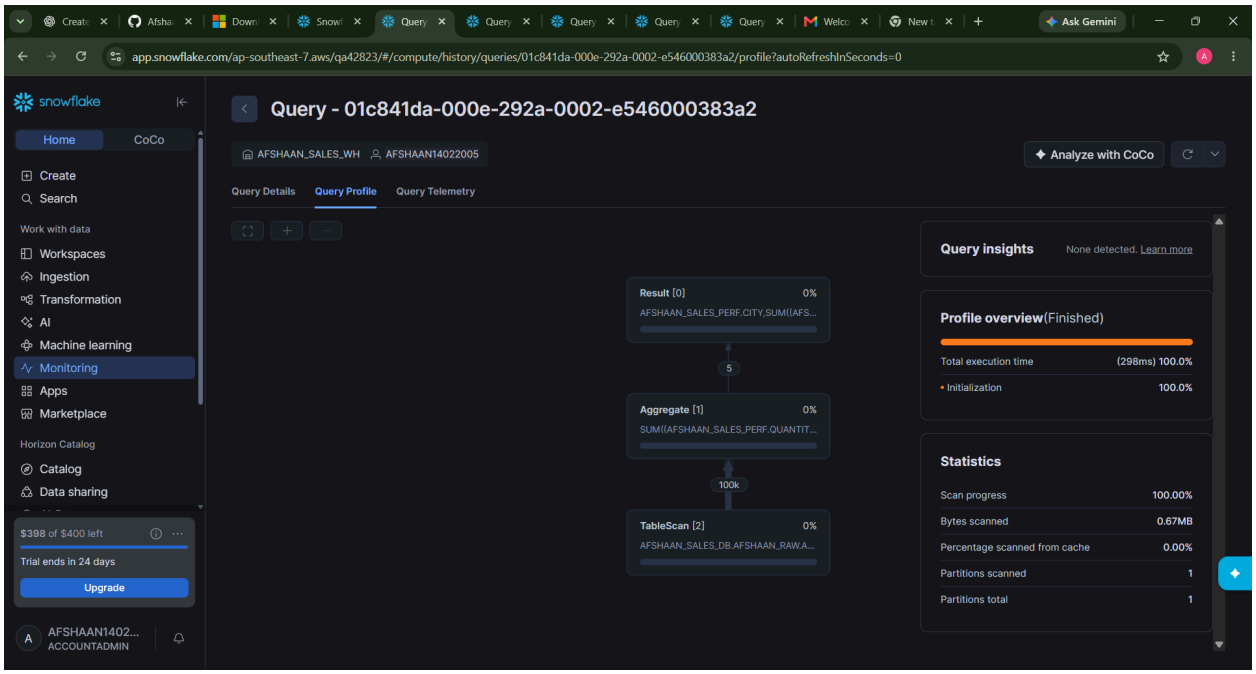
OPTIMIZED

Exclude records with NULL Values

```
SELECT
    CITY,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE,
    COUNT(*) AS TOTAL_ORDERS
```

```
FROM AFSHAAN_SALES_PERF
WHERE CITY IS NOT NULL
GROUP BY CITY;
```

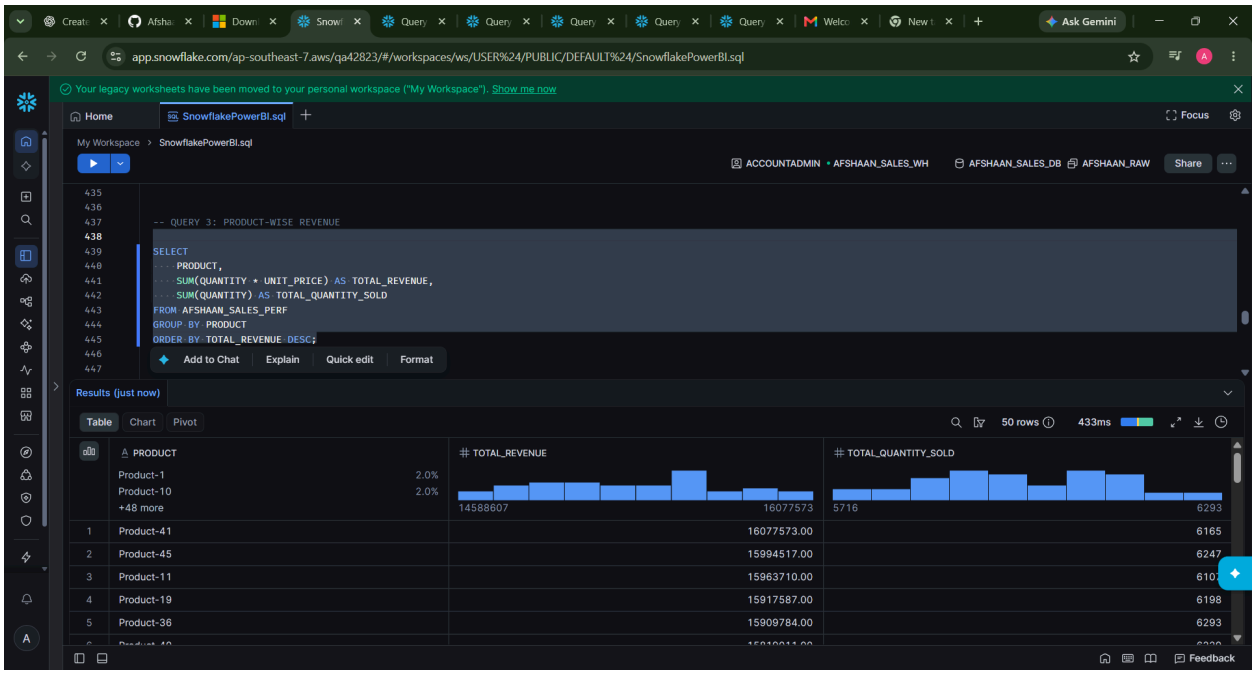


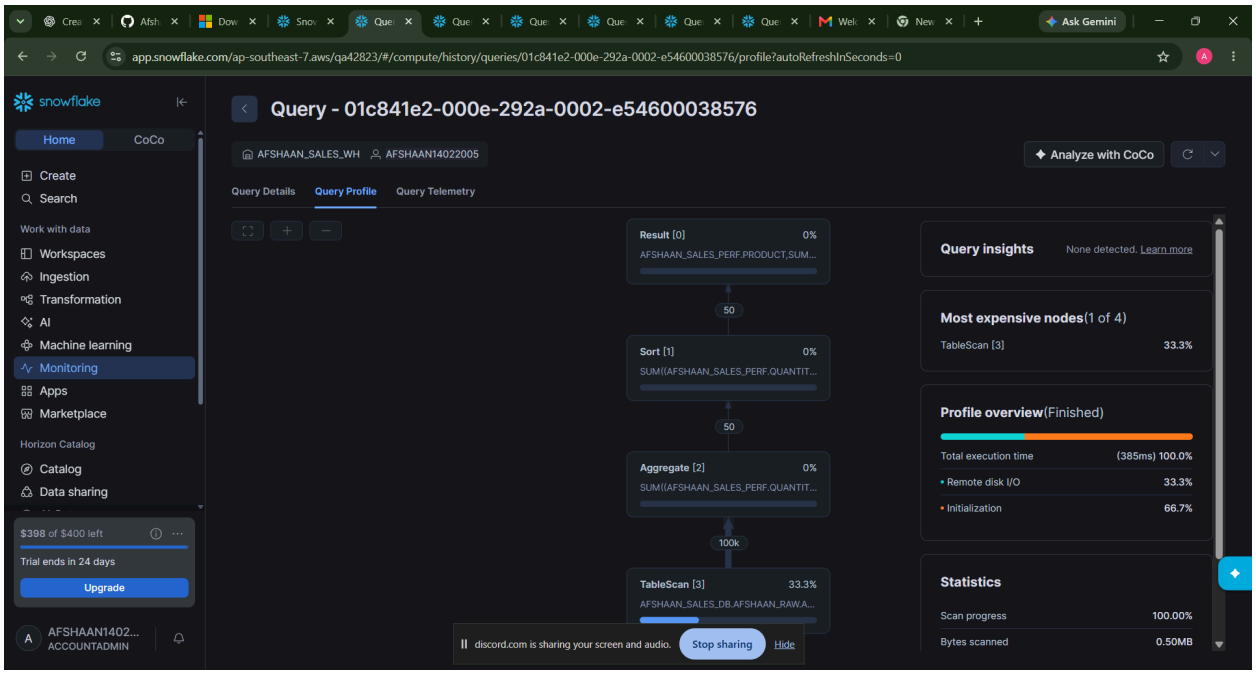


QUERY 3: PRODUCT-WISE REVENUE
BASELINE

```
SELECT
    PRODUCT,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE,
```

```
    SUM(QUANTITY) AS TOTAL_QUANTITY_SOLD
FROM AFSHAAN_SALES_PERF
GROUP BY PRODUCT
ORDER BY TOTAL_REVENUE DESC;
```





OPTIMIZED

Calculate revenue using CTE

```
WITH SALES_CALCULATED AS (

    SELECT
        PRODUCT,
        QUANTITY,
        QUANTITY * UNIT_PRICE AS REVENUE

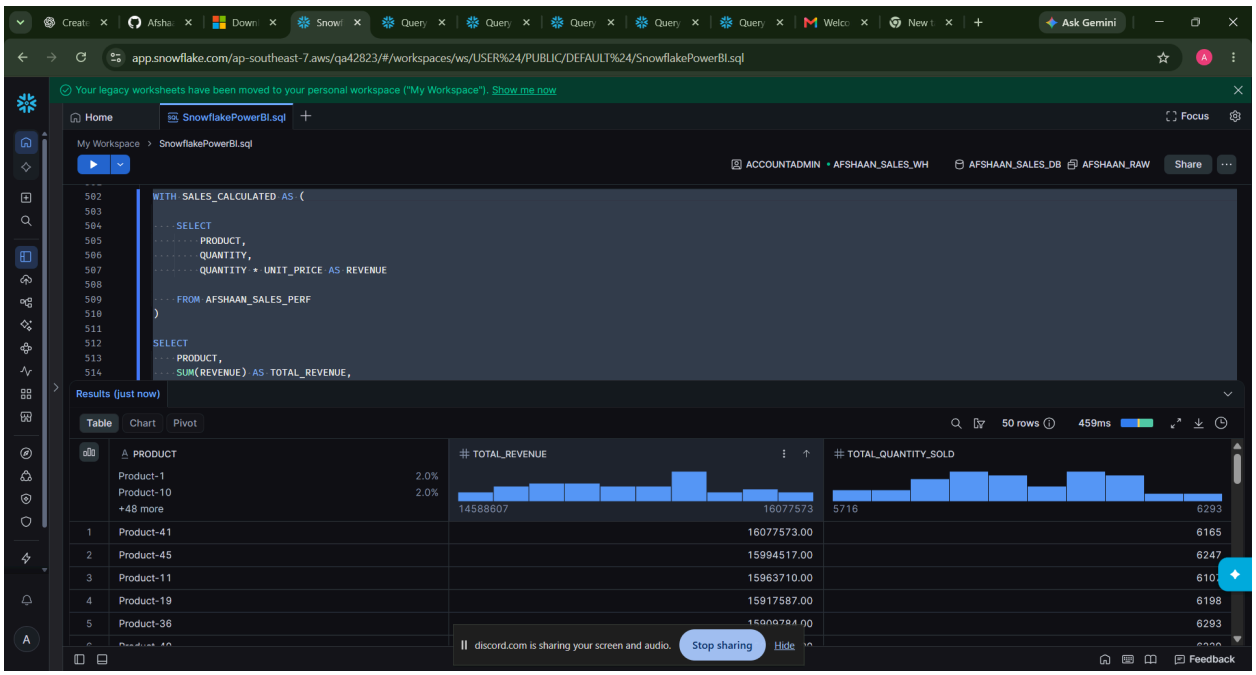
    FROM AFSHAAN_SALES_PERF
)

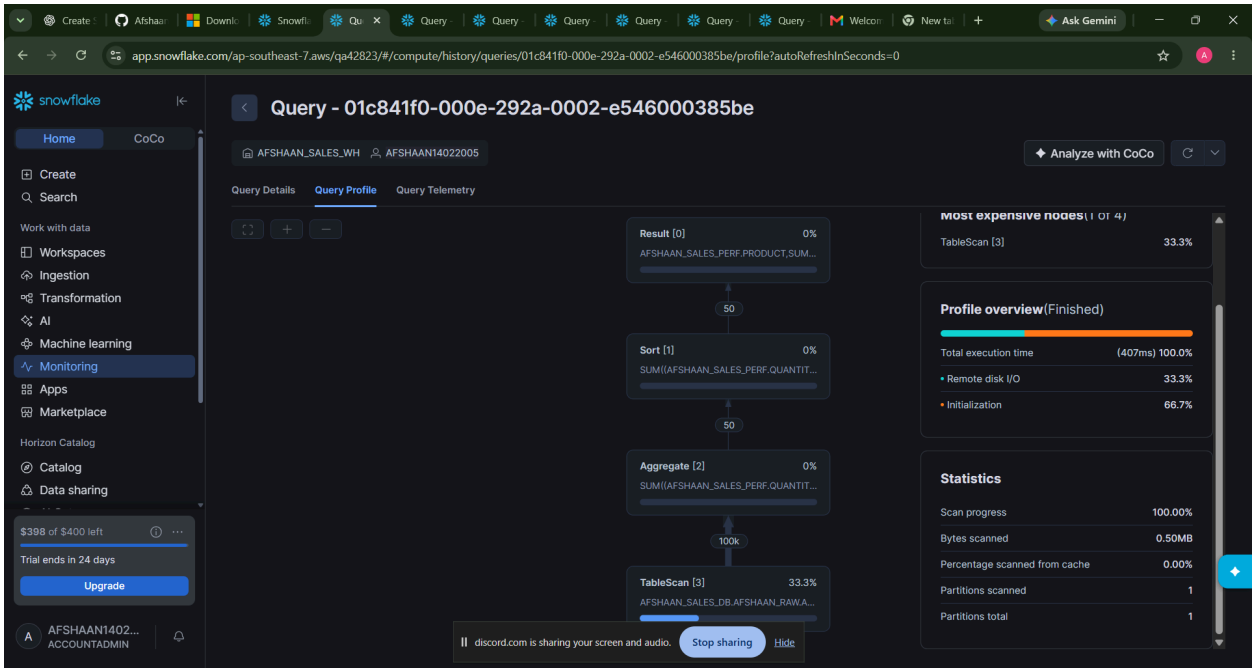
SELECT
    PRODUCT,
    SUM(REVENUE) AS TOTAL_REVENUE,
    SUM(QUANTITY) AS TOTAL_QUANTITY_SOLD

FROM SALES_CALCULATED

GROUP BY PRODUCT

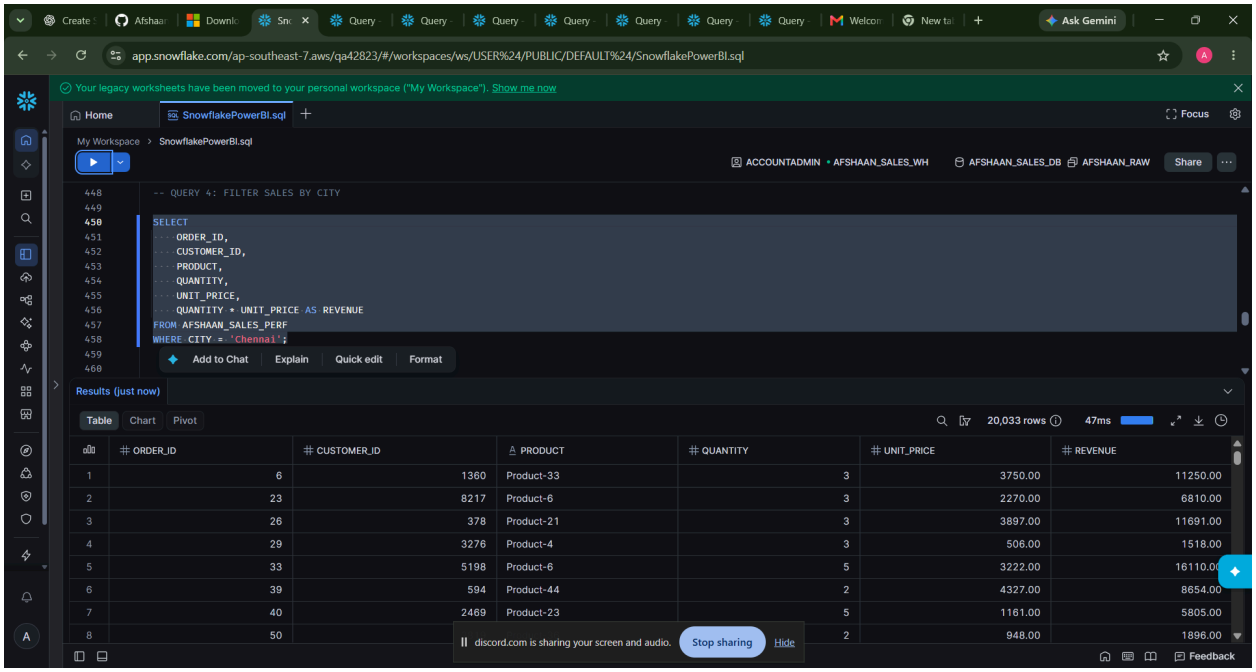
ORDER BY TOTAL_REVENUE DESC;
```

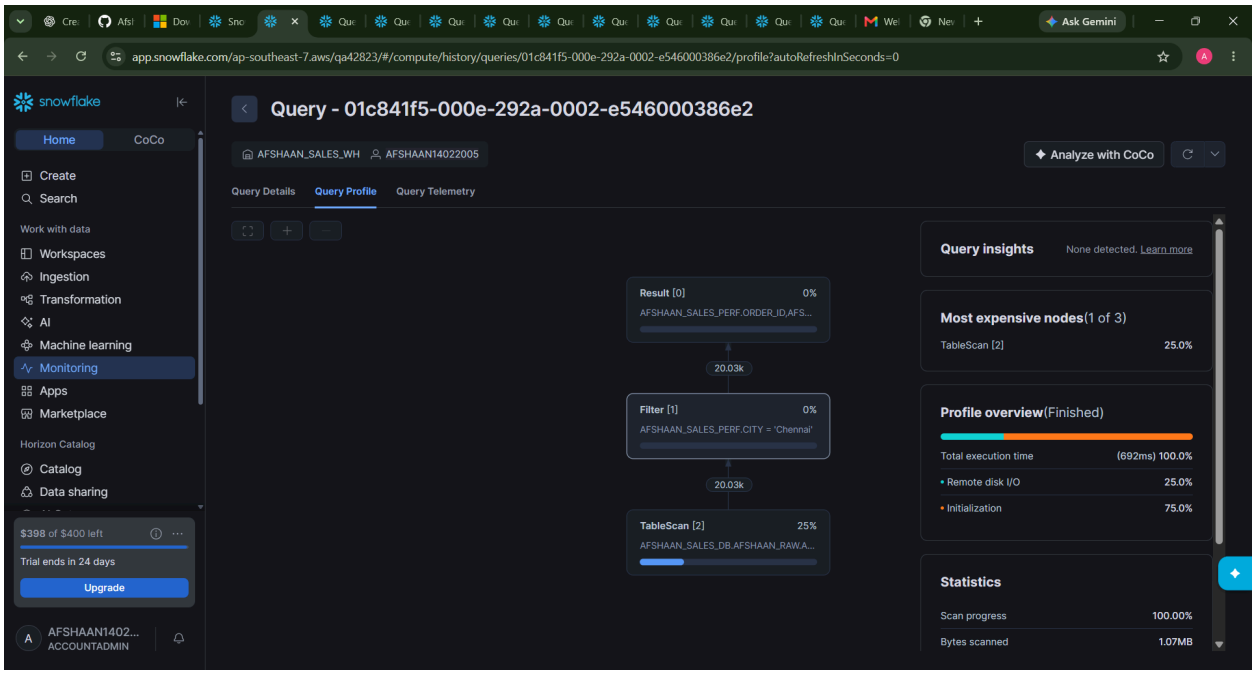




QUERY 4: FILTER SALES BY CITY
BASELINE

```
SELECT
    ORDER_ID,
    CUSTOMER_ID,
    PRODUCT,
    QUANTITY,
    UNIT_PRICE,
    QUANTITY * UNIT_PRICE AS REVENUE
FROM AFSHAAN_SALES_PERF
WHERE CITY = 'Chennai';
```

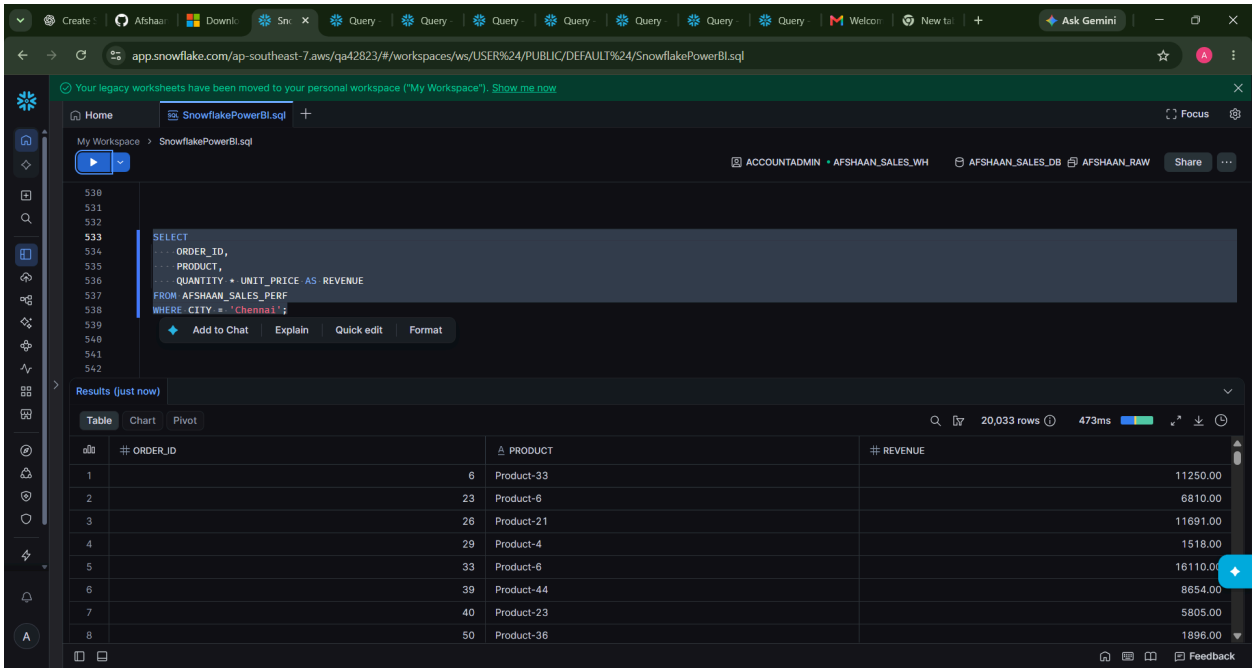


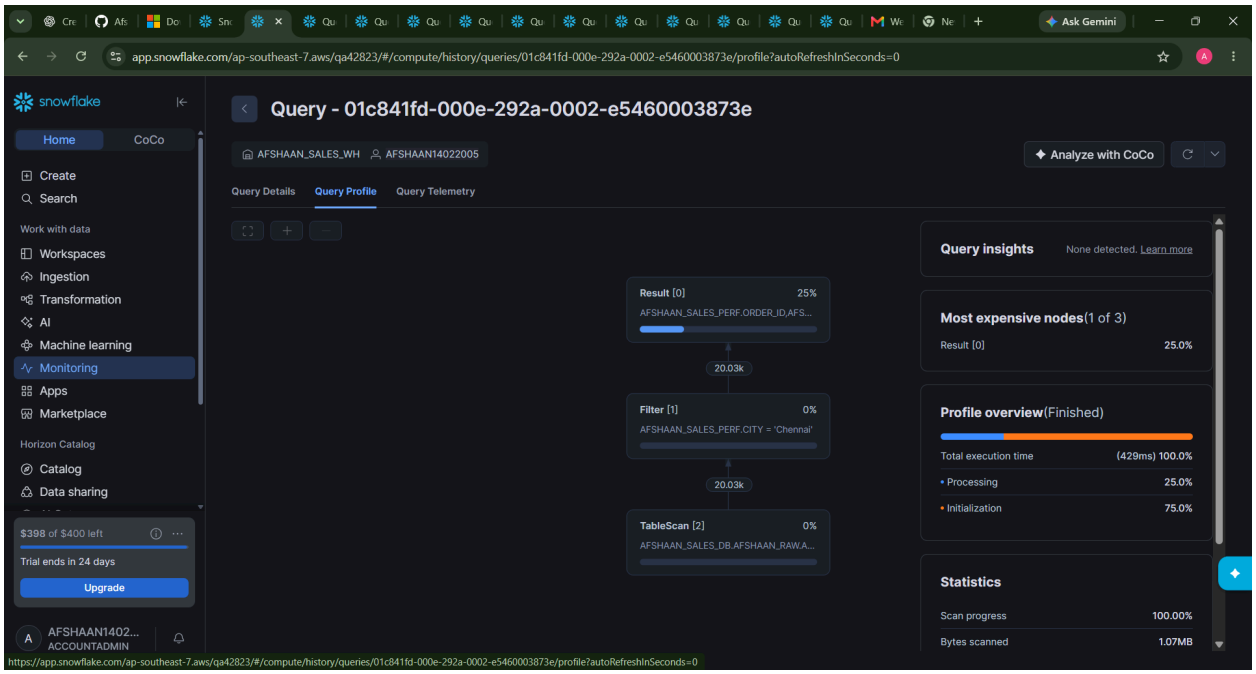


OPTIMIZED

Retrieve only required columns

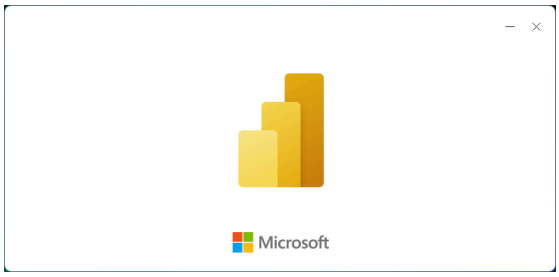
```
SELECT
    ORDER_ID,
    PRODUCT,
    QUANTITY * UNIT_PRICE AS REVENUE
FROM AFSHAAN_SALES_PERF
WHERE CITY = 'Chennai';
```

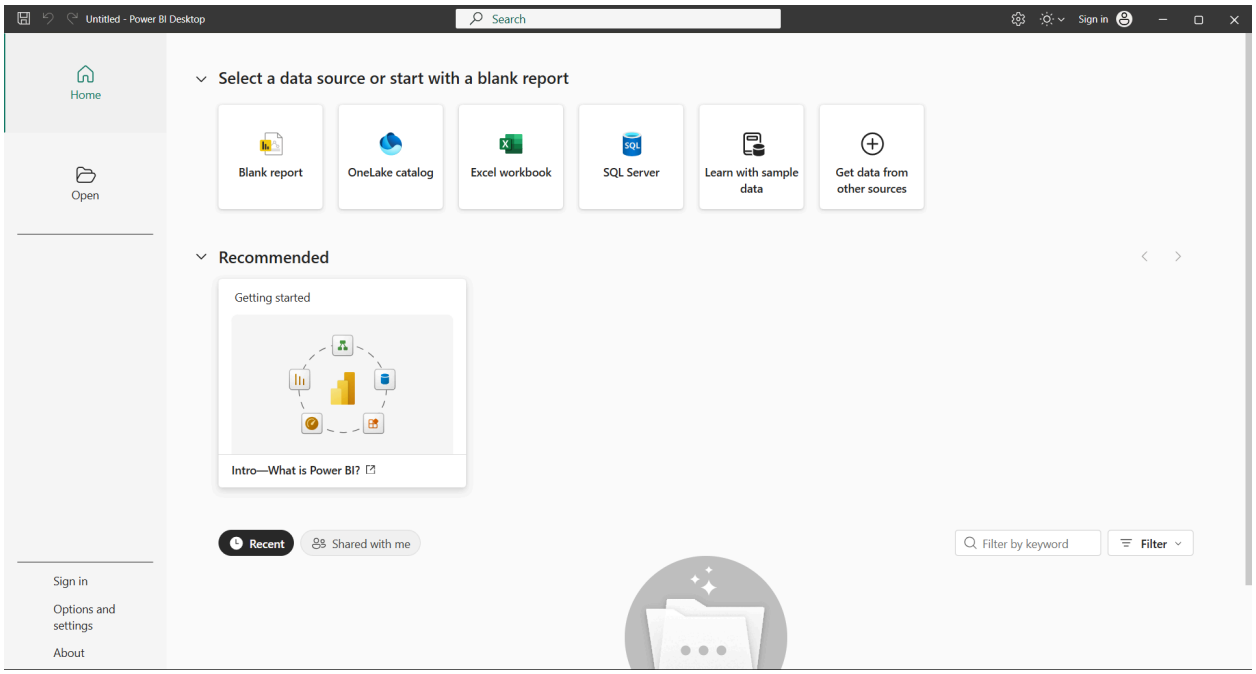

app.snowflake.com/ap-southeast-7.aws/qa42823/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/SnowflakePowerBI.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Home
SnowflakePowerBI.sql
Focus
My Workspace › SnowflakePowerBI.sql
ACCOUNTADMIN • AFSHAAN_SALES_WH
AFSHAAN_SALES_DB AFSHAAN_RAW
Share
SELECT
ORDER_ID,
PRODUCT,
QUANTITY * UNIT_PRICE AS REVENUE
FROM AFSHAAN_SALES_PERF
WHERE CITY = 'Chennai';
Add to Chat
Explain
Quick edit
Format
Results (just now)
Table
Chart
Pivot
20,033 rows
473ms
ORDER_ID
PRODUCT
REVENUE
1 6 Product-33 11250.00
2 23 Product-6 6810.00
3 26 Product-21 11691.00
4 29 Product-4 1518.00
5 33 Product-6 16110.00
6 39 Product-44 8654.00
7 40 Product-23 5805.00
8 50 Product-36 1896.00
Feedback


app.snowflake.com/ap-southeast-7.aws/qa42823/#/compute/history/queries/01c841fd-000e-292a-0002-e5460003873e/profile?autoRefreshInSeconds=0
snowflake
Home
CoCo
Create
Search
Work with data
Workspaces
Ingestion
Transformation
AI
Machine learning
Monitoring
Apps
Marketplace
Horizon Catalog
Catalog
Data sharing
$398 of $400 left
Trial ends in 24 days
Upgrade
AFSHAAN1402...
ACCOUNTADMIN
Query - 01c841fd-000e-292a-0002-e5460003873e
AFSHAAN_SALES_WH
AFSHAAN14022005
Analyze with CoCo
Query Details
Query Profile
Query Telemetry
Result [0] 25%
AFSHAAN_SALES_PERF.ORDER_ID,AFS...
20.03k
Filter [1] 0%
AFSHAAN_SALES_PERF.CITY = 'Chennai'
20.03k
TableScan [2] 0%
AFSHAAN_SALES_DB.AFSHAAN_RAW.A...
Query insights
None detected. Learn more
Most expensive nodes(1 of 3)
Result [0] 25.0%
Profile overview(Finished)
Total execution time (429ms) 100.0%
Processing 25.0%
Initialization 75.0%
Statistics
Scan progress 100.00%
Bytes scanned 1.07MB

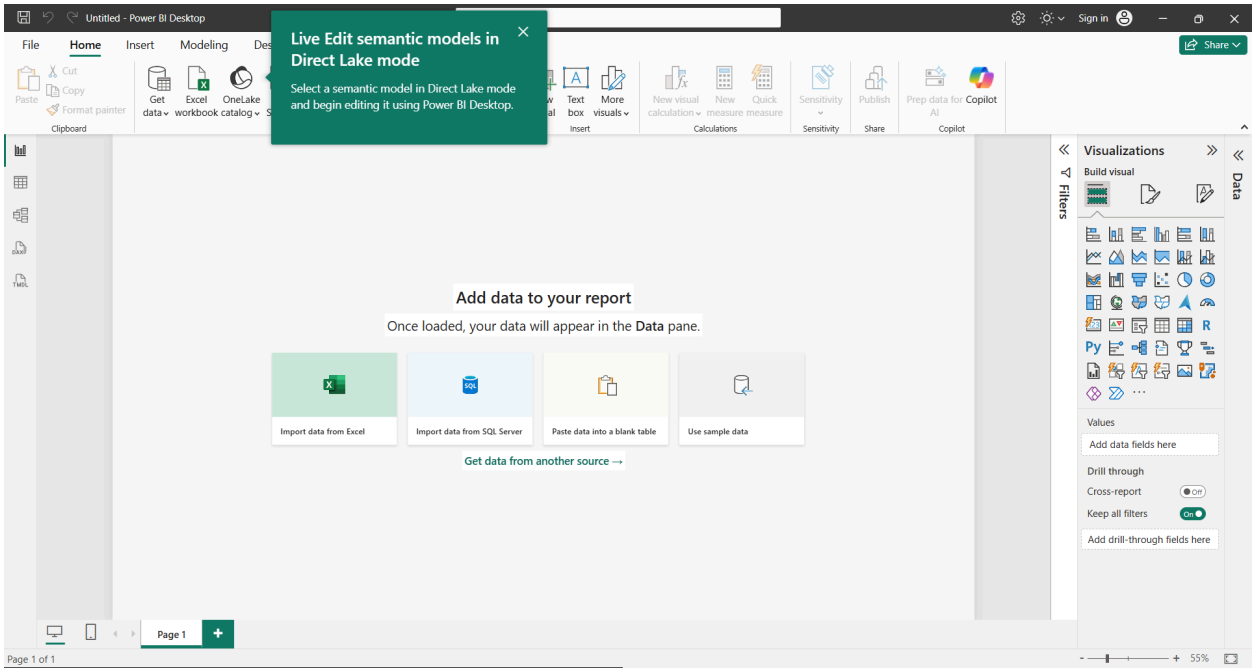
POWER BI

Download and open PowerBI

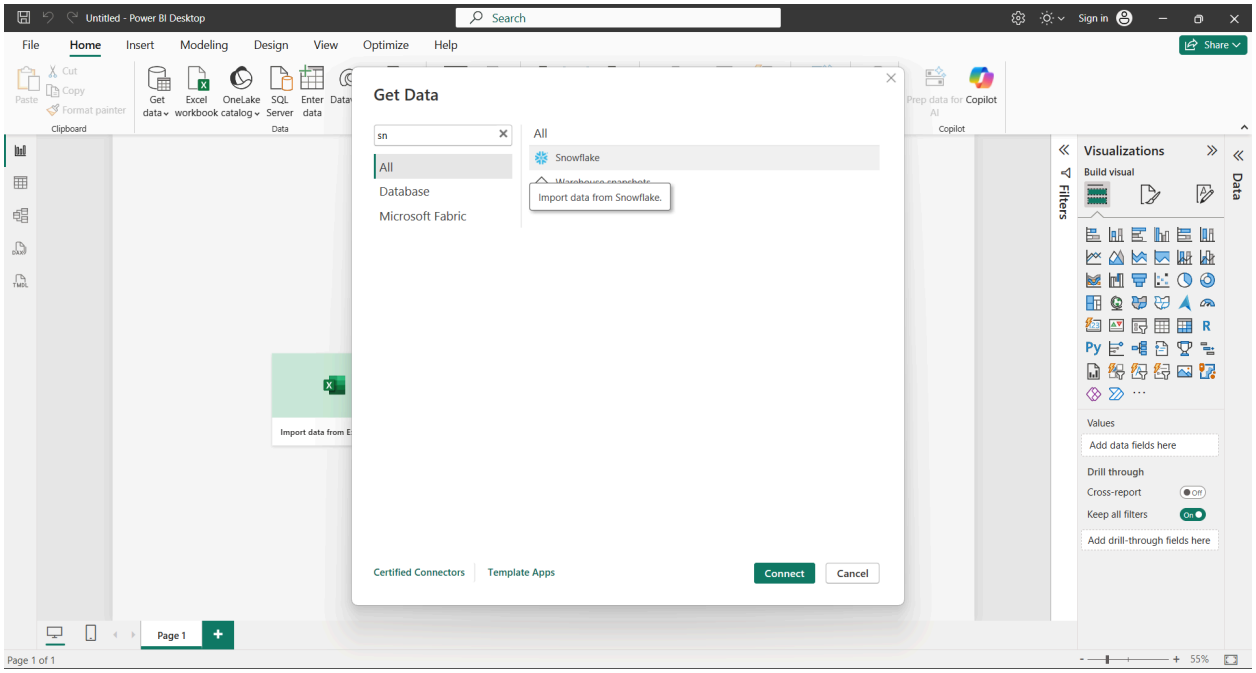




Click Blank Report



Get data from other sources → Snowflake

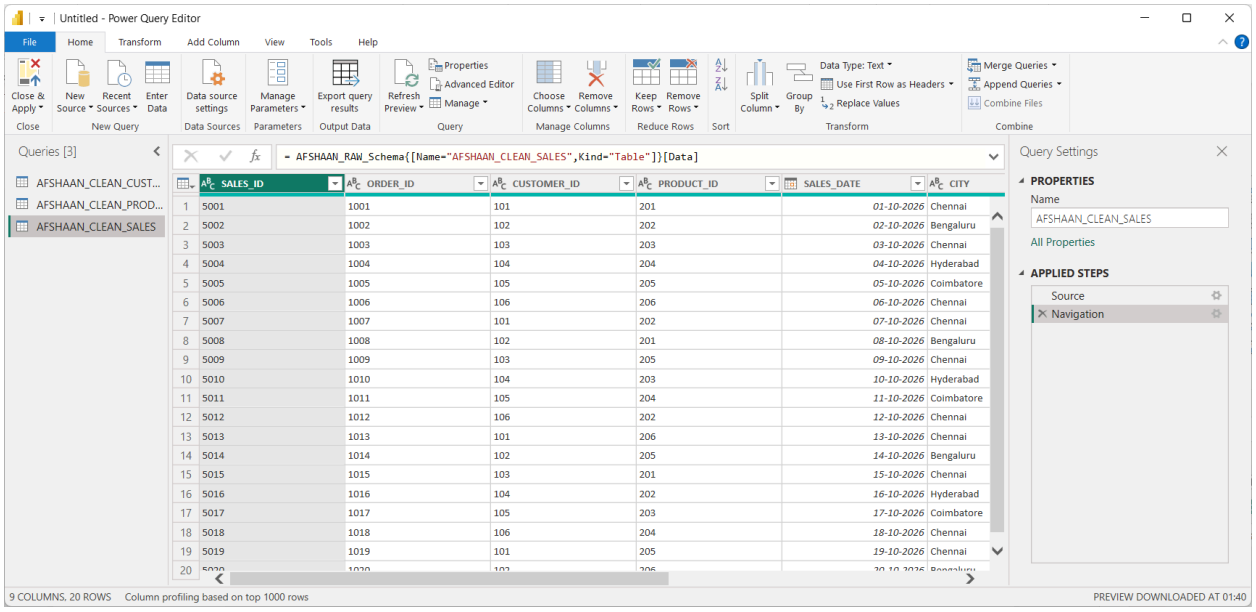


Fill the URL and the warehouse name

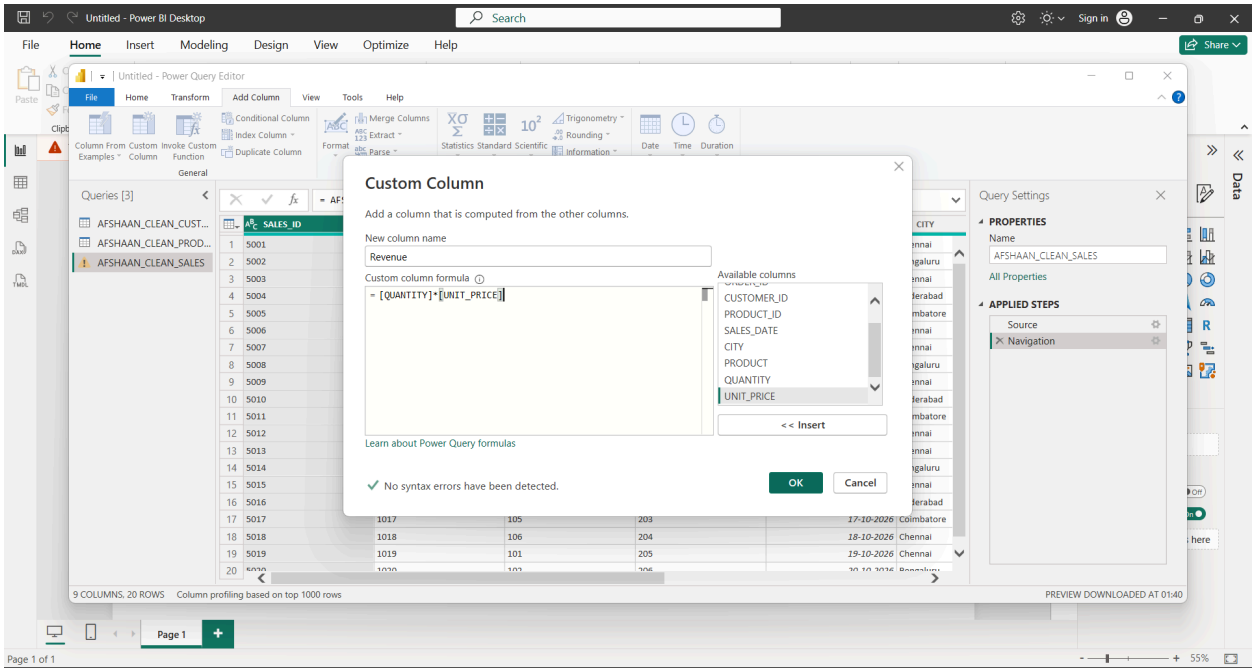


Enter Username and Password of your Snowflake Account

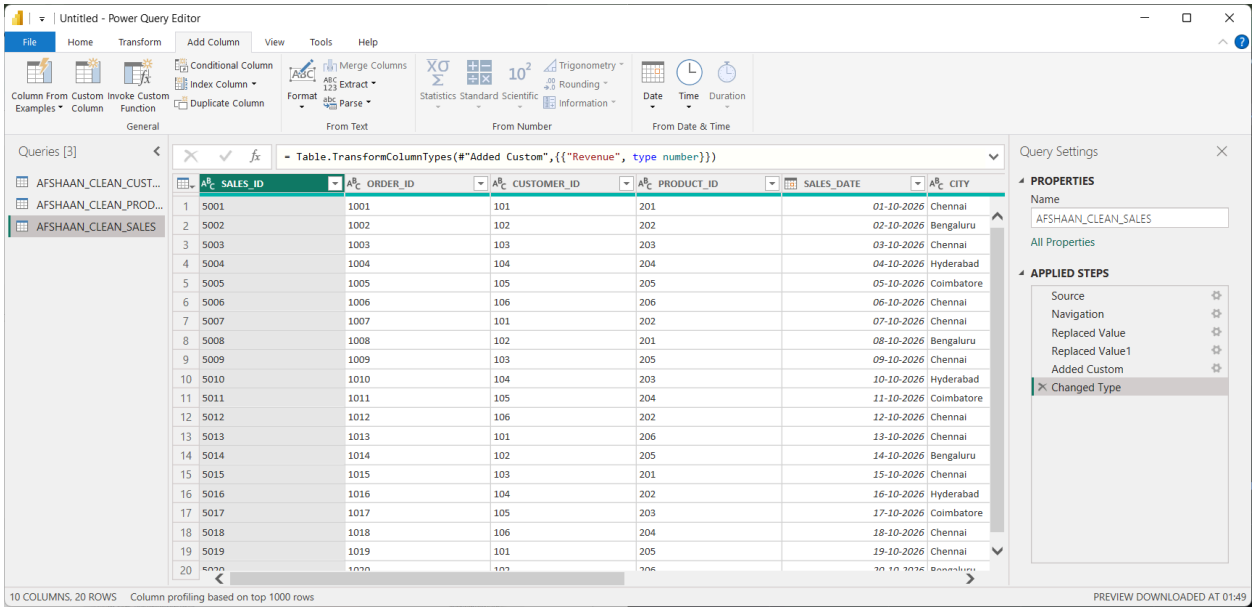
Select the tables that you need to load and click →Transform data



Add Column

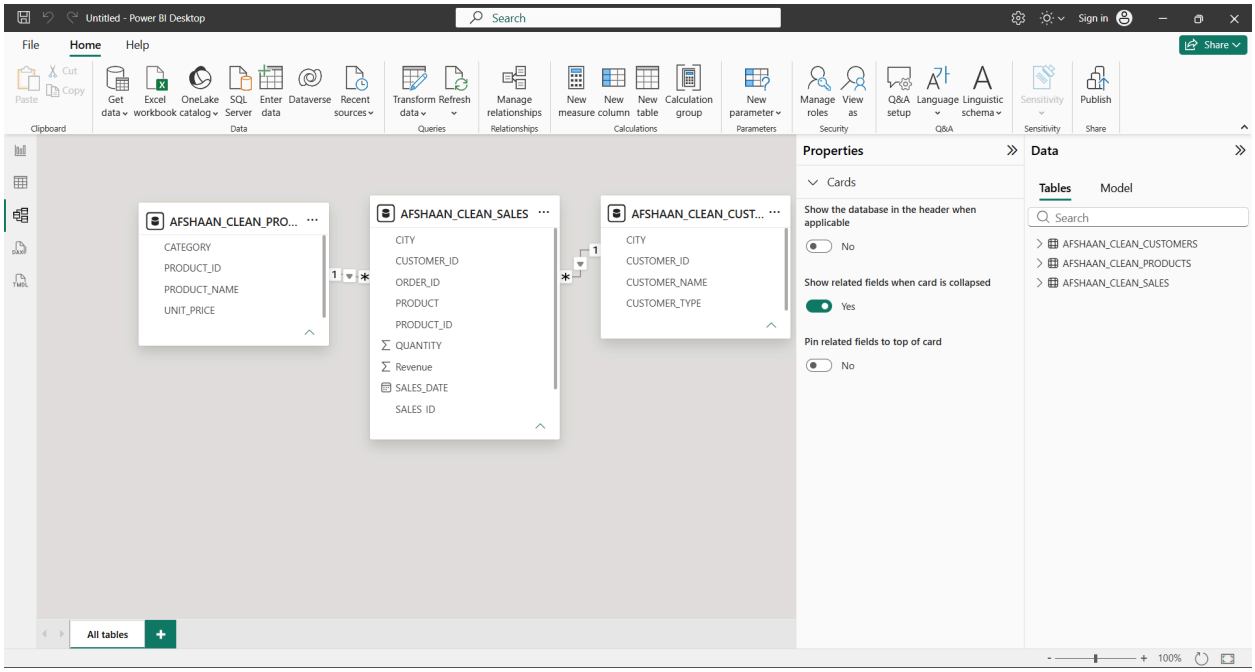


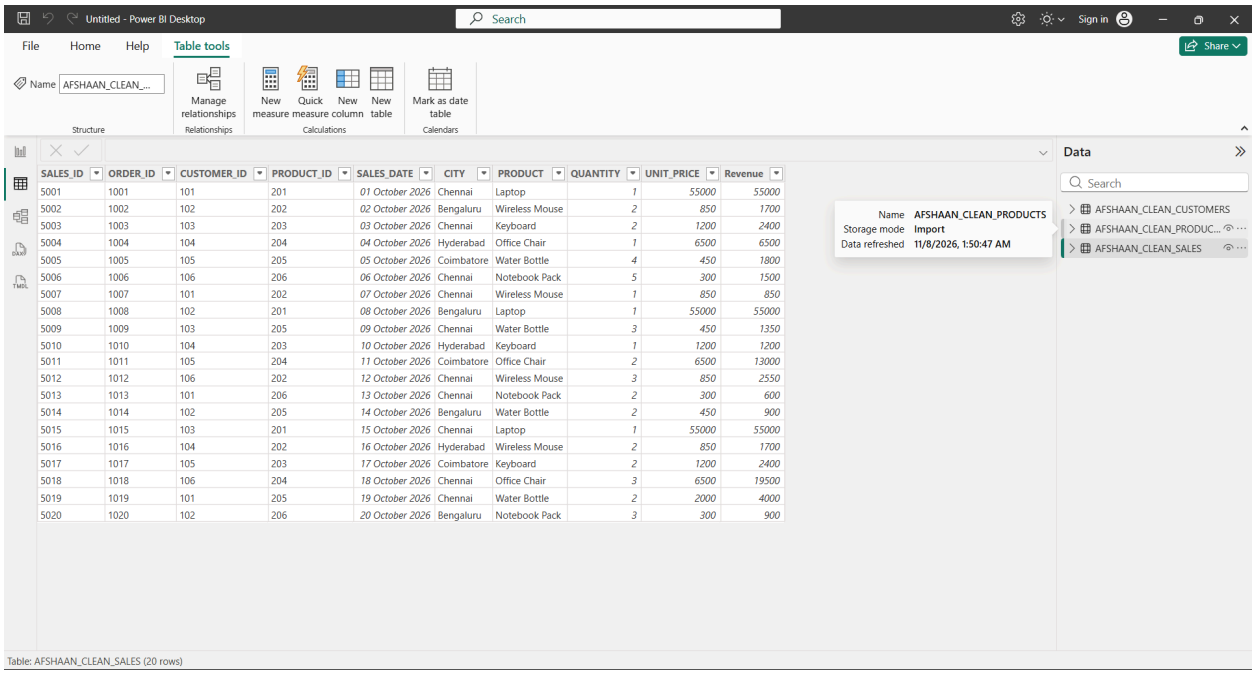
Check for types and whether its clean or make it appropriate for analysis



Home→Close and apply to Load the data

Add relationships if not already present





SALES_ID	ORDER_ID	CUSTOMER_ID	PRODUCT_ID	SALES_DATE	CITY	PRODUCT	QUANTITY	UNIT_PRICE	Revenue
5001	1001	101	201	01 October 2026	Chennai	Laptop	1	55000	55000
5002	1002	102	202	02 October 2026	Bengaluru	Wireless Mouse	2	850	1700
5003	1003	103	203	03 October 2026	Chennai	Keyboard	2	1200	2400
5004	1004	104	204	04 October 2026	Hyderabad	Office Chair	1	6500	6500
5005	1005	105	205	05 October 2026	Coimbatore	Water Bottle	4	450	1800
5006	1006	106	206	06 October 2026	Chennai	Notebook Pack	5	300	1500
5007	1007	101	202	07 October 2026	Chennai	Wireless Mouse	1	850	850
5008	1008	102	201	08 October 2026	Bengaluru	Laptop	1	55000	55000
5009	1009	103	205	09 October 2026	Chennai	Water Bottle	3	450	1350
5010	1010	104	203	10 October 2026	Hyderabad	Keyboard	1	1200	1200
5011	1011	105	204	11 October 2026	Coimbatore	Office Chair	2	6500	13000
5012	1012	106	202	12 October 2026	Chennai	Wireless Mouse	3	850	2550
5013	1013	101	206	13 October 2026	Chennai	Notebook Pack	2	300	600
5014	1014	102	205	14 October 2026	Bengaluru	Water Bottle	2	450	900
5015	1015	103	201	15 October 2026	Chennai	Laptop	1	55000	55000
5016	1016	104	202	16 October 2026	Hyderabad	Wireless Mouse	2	850	1700
5017	1017	105	203	17 October 2026	Coimbatore	Keyboard	2	1200	2400
5018	1018	106	204	18 October 2026	Chennai	Office Chair	3	6500	19500
5019	1019	101	205	19 October 2026	Chennai	Water Bottle	2	2000	4000
5020	1020	102	206	20 October 2026	Bengaluru	Notebook Pack	3	300	900

Create 4 new measures

Total Revenue =SUM(SALES_DATA[Sales Amount])

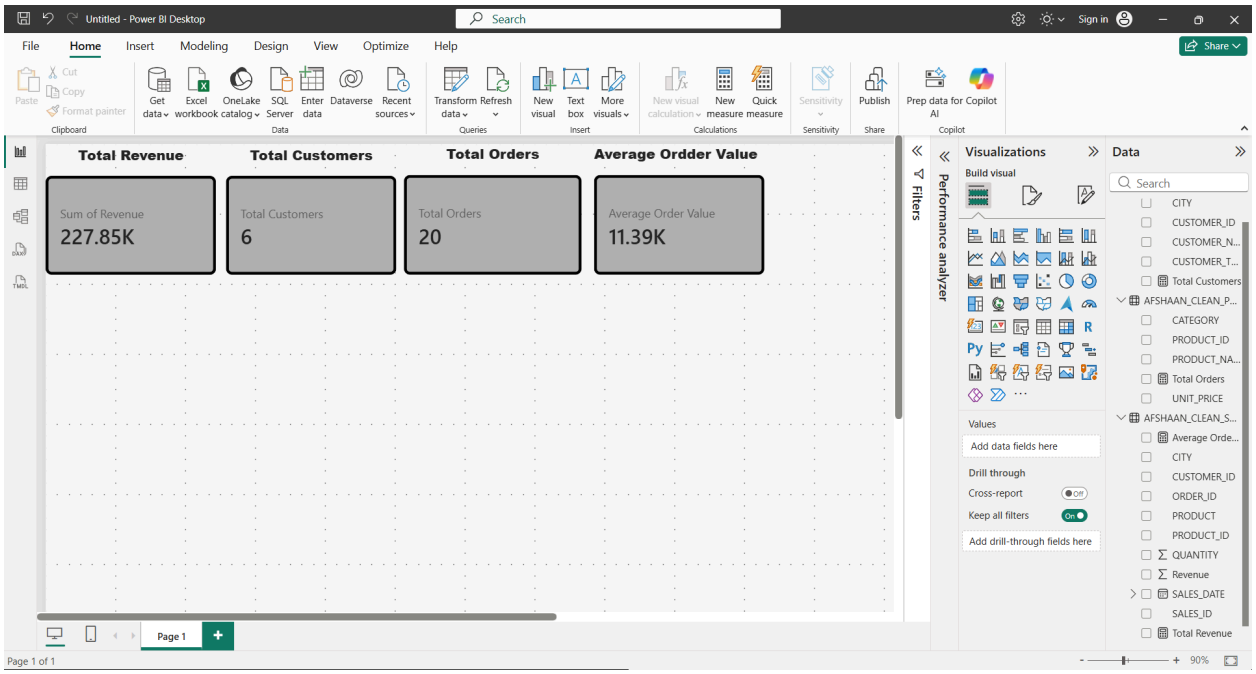
Total Orders =DISTINCTCOUNT(SALES_DATA[ORDER_ID])

Total Customers =DISTINCTCOUNT(SALES_DATA[CUSTOMER_ID])

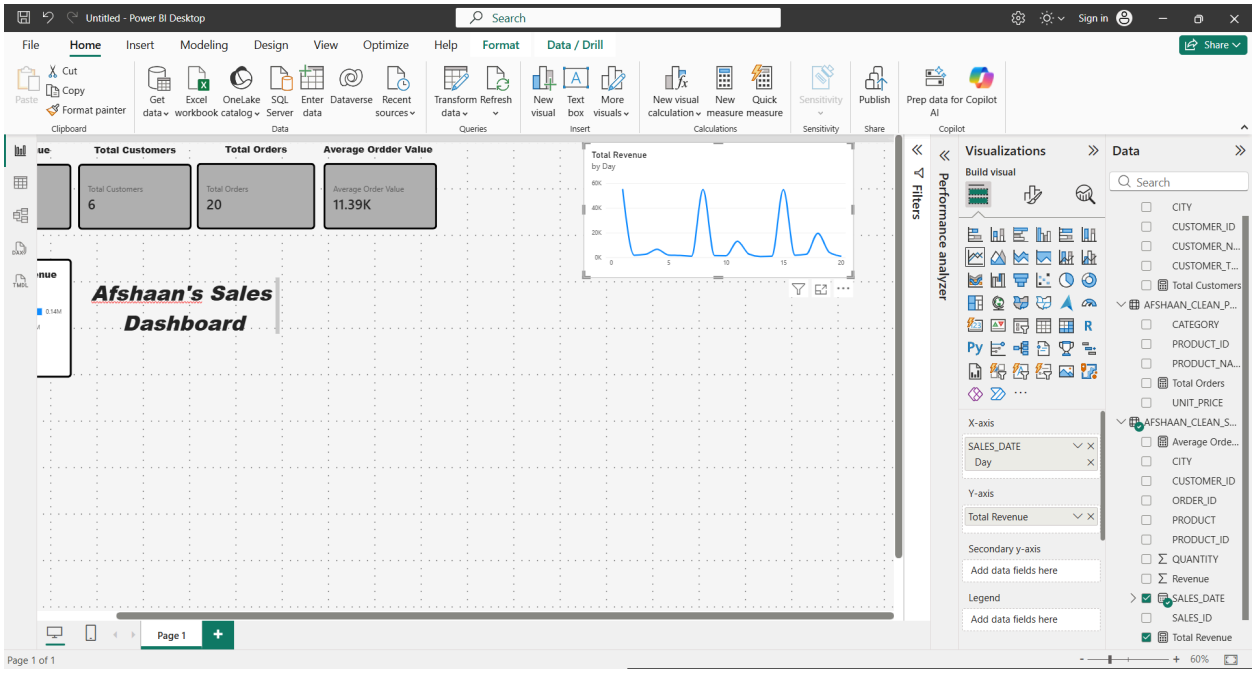
Average Order Value =DIVIDE([Total Revenue], [Total Orders], 0)

Create Cards

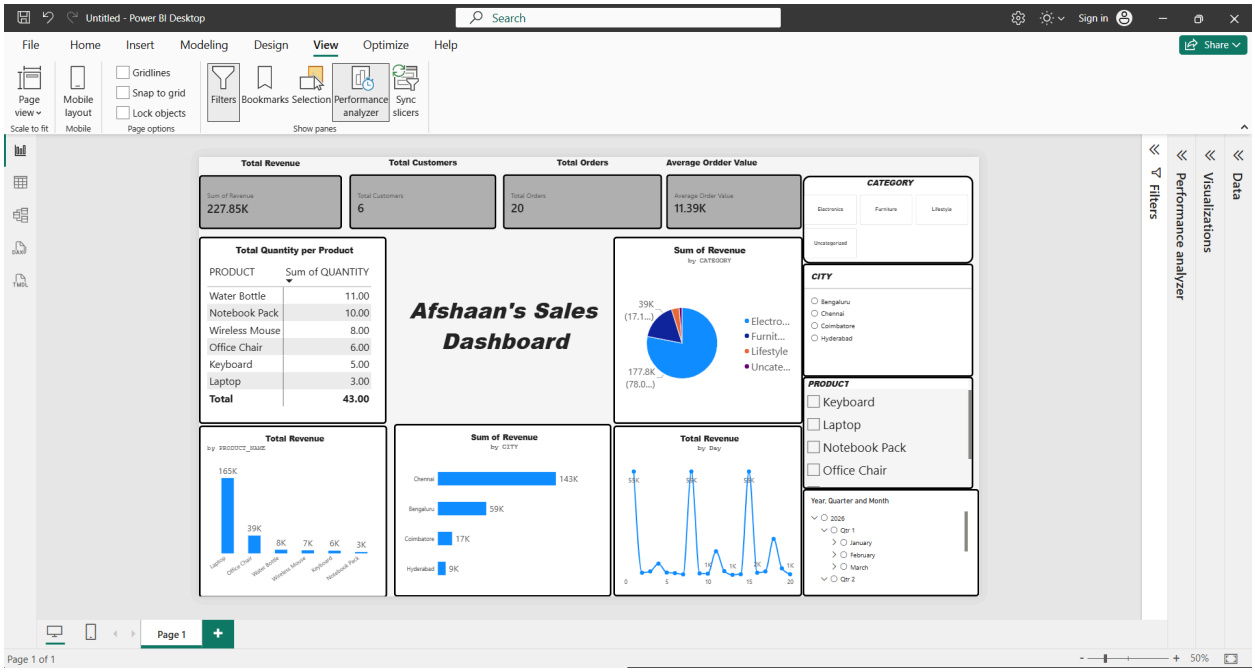
Select these measures and edit visuals by going to Format visuals

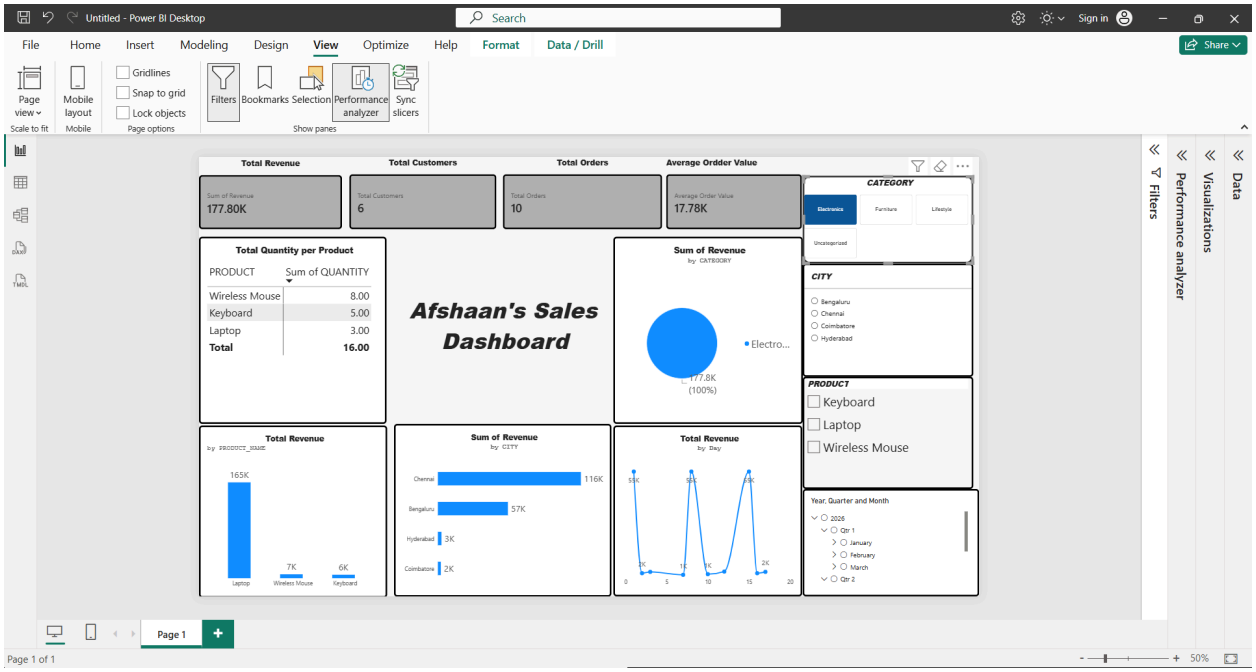


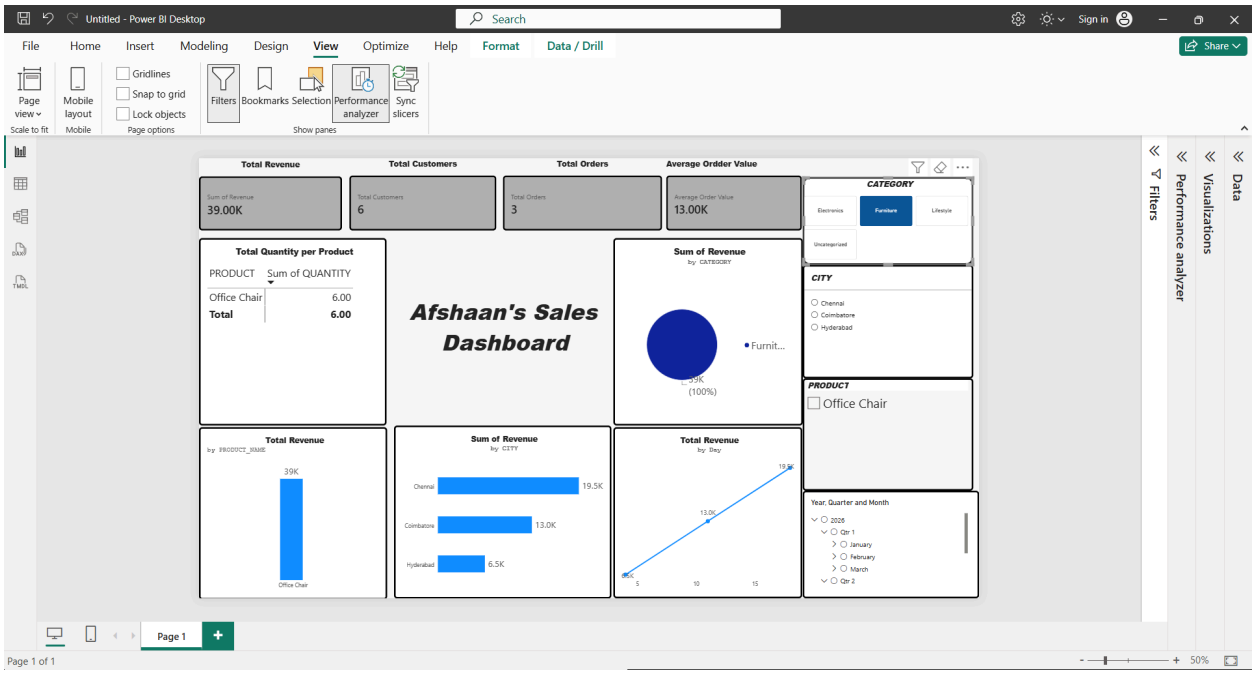
Create charts and graphs

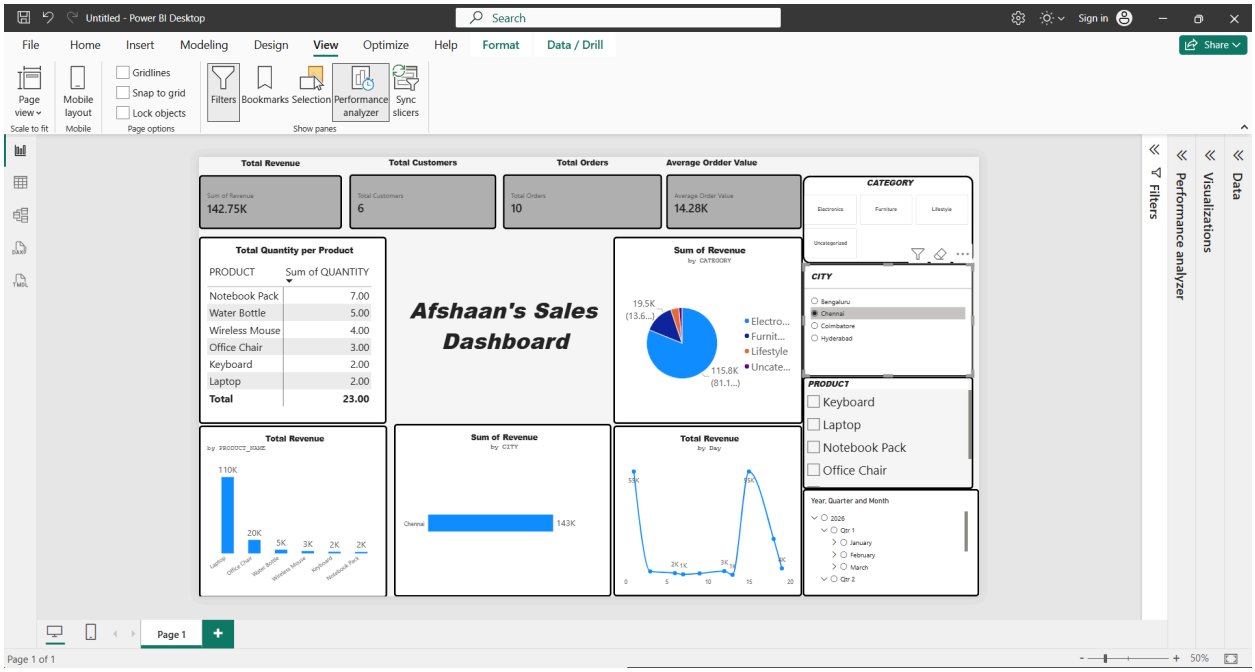


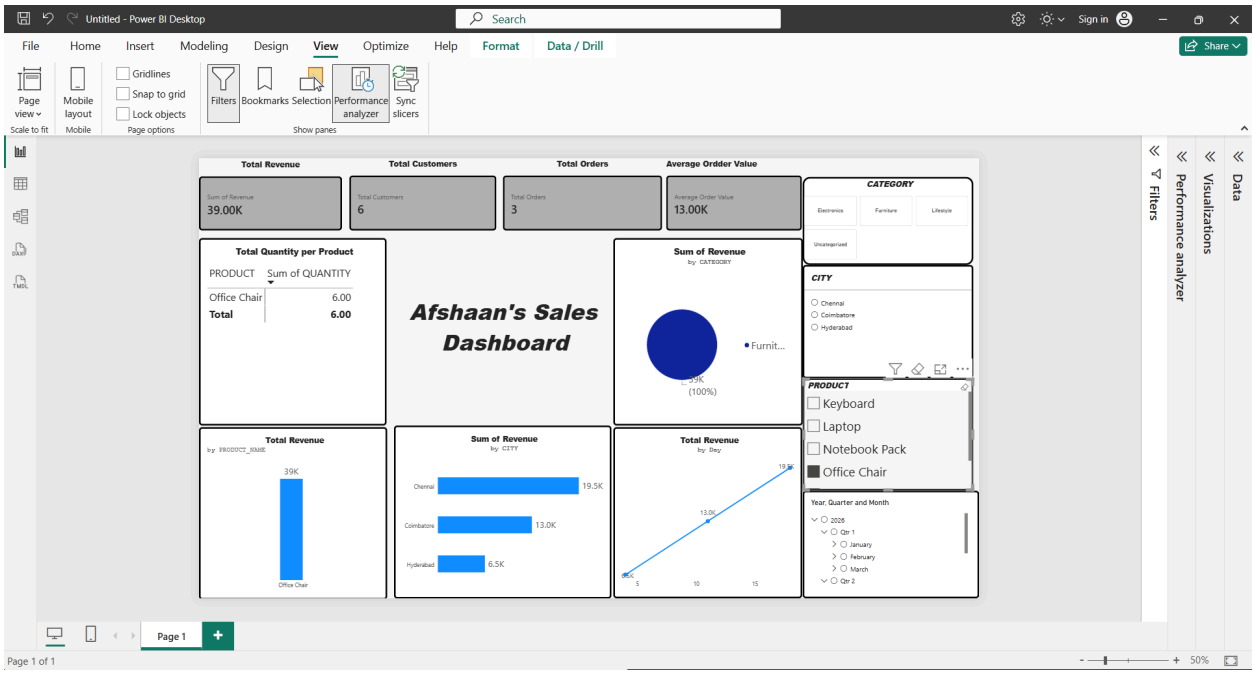
Create Slicers

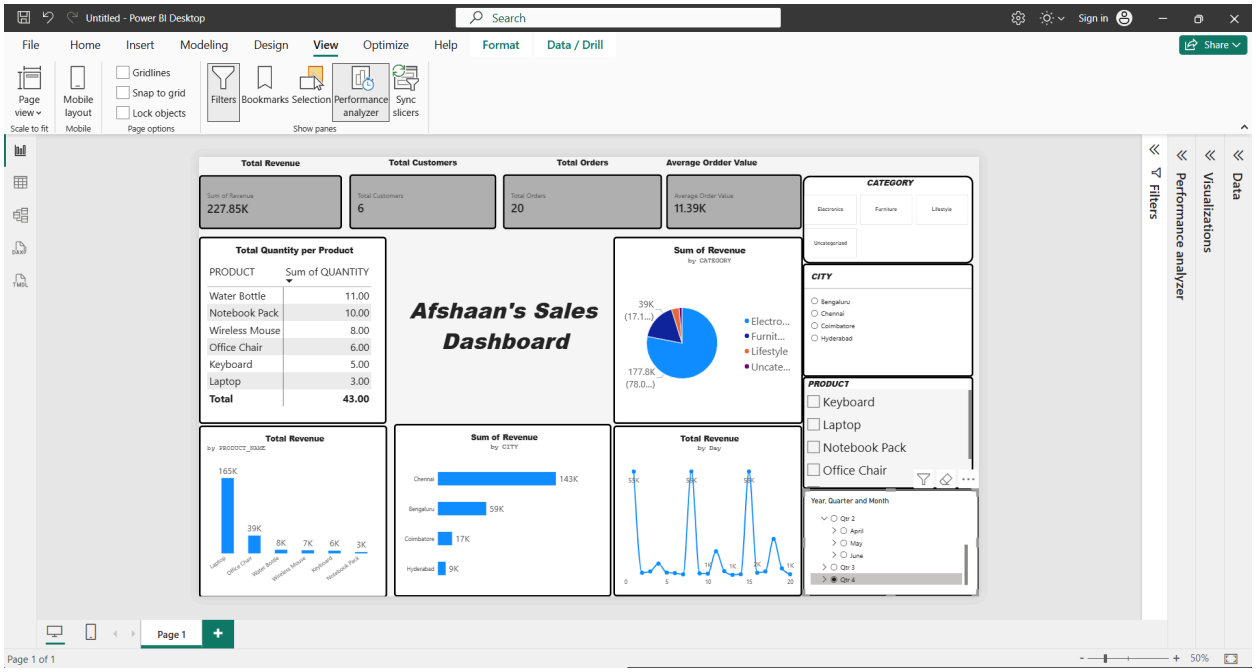

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Total Revenue
Sum of Revenue
227.85K
Total Customers
Total Customers
6
Total Orders
Total Orders
20
Average Ordder Value
Average Order Value
11.39K
CATEGORY
Electronics
Furniture
Lifestyle
Uncategorized
Total Quantity per Product
PRODUCT
Sum of QUANTITY
Water Bottle 11.00
Notebook Pack 10.00
Wireless Mouse 8.00
Office Chair 6.00
Keyboard 5.00
Laptop 3.00
Total 43.00
Afshaan's Sales Dashboard
Sum of Revenue
by CATEGORY
39K
(17.1...)
177.8K
(78.0...)
Electro...
Furnit...
Lifestyle
Uncate...
CITY
Bengaluru
Chennai
Coimbatore
Hyderabad
PRODUCT
Keyboard
Laptop
Notebook Pack
Office Chair
Total Revenue
by PRODUCT_NAME
165K
39K
8K
7K
6K
3K
Sum of Revenue
by CITY
Chennai 143K
Bengaluru 59K
Coimbatore 17K
Hyderabad 9K
Total Revenue
by Day
Year, Quarter and Month
2026
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March
Qtr 2
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Total Revenue
Total Customers
Total Orders
Average Order Value
177.80K
6
10
17.78K
Total Quantity per Product
PRODUCT
Sum of QUANTITY
Wireless Mouse 8.00
Keyboard 5.00
Laptop 3.00
Total 16.00
Afshaan's Sales Dashboard
Sum of Revenue
by CATEGORY
Electro...
177.8K
(100%)
CATEGORY
Electronics
Furniture
Lifestyle
Uncategorized
CITY
Bengaluru
Chennai
Coimbatore
Hyderabad
PRODUCT
Keyboard
Laptop
Wireless Mouse
Total Revenue
by PRODUCT_NAME
165K
7K
6K
Laptop
Wireless Mouse
Keyboard
Sum of Revenue
by CITY
Chennai 116K
Bengaluru 57K
Hyderabad 3K
Coimbatore 2K
Total Revenue
by Day
Year, Quarter and Month
2026
Qtr 1
January
February
March
Qtr 2
Filters
Performance analyzer
Visualizations
Data
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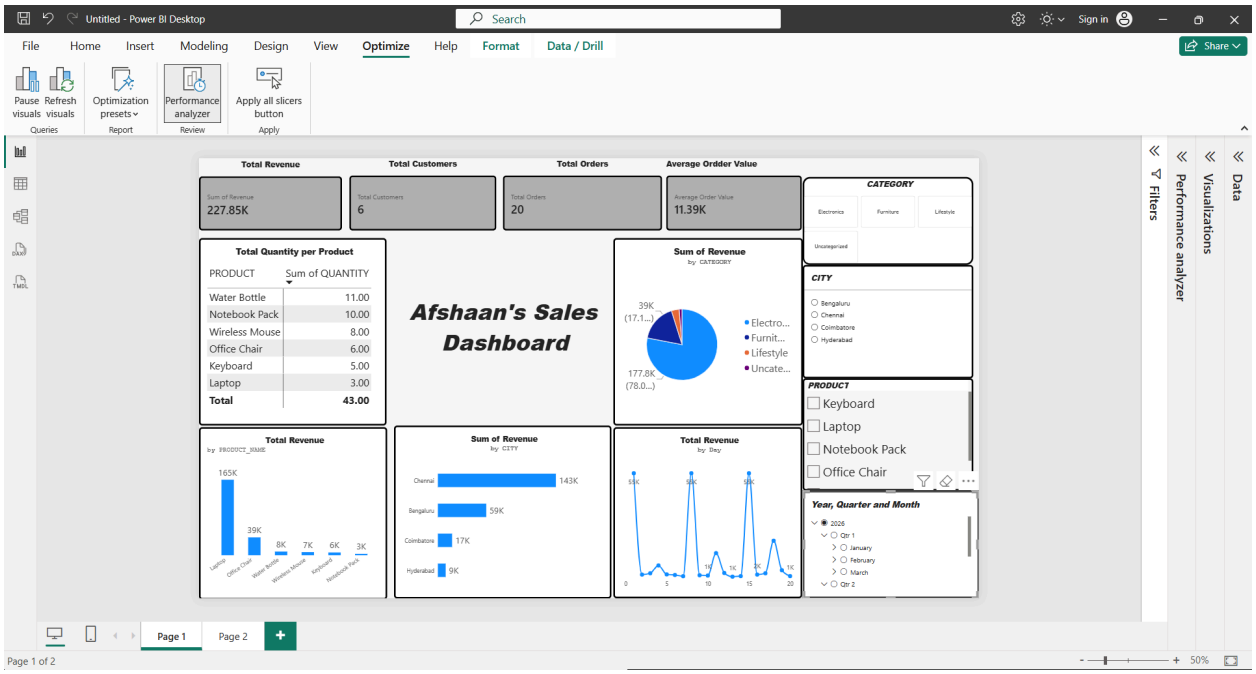

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Total Revenue
Total Customers
Total Orders
Average Order Value
39.00K
6
3
13.00K
Total Quantity per Product
PRODUCT
Sum of QUANTITY
Office Chair 6.00
Total 6.00
Afshaan's Sales Dashboard
Sum of Revenue
by CATEGORY
Furnit...
39K
(100%)
CATEGORY
Electronics
Furniture
Lifestyle
Uncategorized
CITY
Chennai
Coimbatore
Hyderabad
PRODUCT
Office Chair
Total Revenue
by PRODUCT_NAME
39K
Office Chair
Sum of Revenue
by CITY
Chennai 19.5K
Coimbatore 13.0K
Hyderabad 6.5K
Total Revenue
by Day
13.0K
Year, Quarter and Month
2026
Qtr 1
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Qtr 2
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Selection
Performance analyzer
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Scale to fit
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Page options
Show panes
Total Revenue
Total Customers
Total Orders
Average Ordder Value
142.75K
6
10
14.28K
Total Quantity per Product
PRODUCT
Sum of QUANTITY
Notebook Pack 7.00
Water Bottle 5.00
Wireless Mouse 4.00
Office Chair 3.00
Keyboard 2.00
Laptop 2.00
Total 23.00
Afshaan's Sales Dashboard
Sum of Revenue
by CATEGORY
Electro...
Furnit...
Lifestyle
Uncate...
CATEGORY
CITY
Bengaluru
Chennai
Coimbatore
Hyderabad
PRODUCT
Keyboard
Laptop
Notebook Pack
Office Chair
Total Revenue
by PRODUCT_NAME
Sum of Revenue
by CITY
Chennai
143K
Total Revenue
by Day
Year, Quarter and Month
2026
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Chennai
Coimbatore
Hyderabad
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Laptop
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Office Chair
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19.5K
Coimbatore
13.0K
Hyderabad
6.5K
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Total Revenue
by PRODUCT_NAME
Sum of Revenue
by CITY
Total Revenue
by Day
CATEGORY
CITY
PRODUCT
Keyboard
Laptop
Notebook Pack
Office Chair
Year, Quarter and Month
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Pause visuals
Refresh visuals
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Performance analyzer
Apply all slicers button
Queries
Report
Review
Apply
Total Revenue
227.85K
Total Customers
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Total Orders
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Average Order Value
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Wireless Mouse 8.00
Office Chair 6.00
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Sum of Revenue
by CATEGORY
Total Revenue
by PRODUCT_NAME
Sum of Revenue
by CITY
Total Revenue
by Day
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CITY
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Notebook Pack
Office Chair
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